Joint MBMP-CMSE and NC-MVDR Technologies for FDA-MIMO SAR Mainlobe NMM-ISRJ Suppression

Hanqing Zhang, Guodong Jin, *Senior Member, IEEE*, Hongbo Zhang, Shilin Niu, Jie Lv, Yuan Cheng, Jing Ding, Zhenyu Guo, Shaohua Ye, and Daiyin Zhu, *Member, IEEE*

***Abstract*—The mainlobe jamming suppression constitutes a persistent challenge in contemporary radar systems. The mainlobe noise multiplication modulation interrupted sampling repeater jamming (NMM-ISRJ) exhibits dual characteristics of deception and dense masking, making it difficult to effectively distinguish the real targets in radar detection. Frequency diverse array multi-input multi-output synthetic aperture radar (FDA-MIMO SAR) utilizes its additional controllable degrees of freedom (DOFs) to effectively distinguish between the real targets and mainlobe jamming in the transmit spatial frequency domain. However, the traditional blocking matrix preprocessing (BMP) and improved BMP (IBMP) methods will cause the desired signal to lose array DOFs, leading to issues such as the mainlobe broadening of the antenna pattern and sidelobe level amplitude rise. To address these limitations, this article proposes a joint multistage BMP covariance matrix symmetric extension (MBMP-CMSE) and norm constrained minimum variance distortionless response (NC-MVDR) to suppress mainlobe NMM-ISRJ for FDA-MIMO SAR. First, the mathematical model of MBMP is derived based on the traditional BMP to suppress the desired signal. Then, the CMSE algorithm is applied to avoid the array DOFs loss. Finally, a robust beamforming is achieved using the NC-MVDR algorithm to suppress the mainlobe NMM-ISRJ. The effectiveness of the joint MBMP-CMSE and NC-MVDR in suppressing mainlobe NMM-ISRJ is validated by integrated experiments, including quantitative analysis, point-like targets, and a distributed scene comparative simulation experiment.**

***Index Terms*—Blocking matrix preprocessing (BMP), frequency diverse array multiple-input multiple-output synthetic aperture radar (FDA-MIMO SAR), intermittent sampling repeater jamming (ISRJ), noise multiplication modulation (NMM).**

Received 29 October 2025; revised 25 January 2026 and 22 March 2026; accepted 6 May 2026. Date of publication 12 May 2026; date of current version 27 May 2026. This work was supported in part by the National Natural Science Foundation of China under Grant 62501274 and Grant 62471221, in part by the Basic Research Program of Jiangsu (Natural Science Foundation of Jiangsu Province) under Grant BK20251396, in part by Guangdong Basic and Applied Basic Research Foundation under Grant 2025A1515011644, in part by Shenzhen Science and Technology Program under Grant JCYJ20230807142000001, in part by the Fundamental Research Funds for the Central Universities under Grant NS2025013, in part by China Postdoctoral Science Foundation under Grant 2025M784331, and in part by Shenzhen Longhua Science and Technology Innovation Special Funding Project (Industrial Sci-Tech Innovation Center of Low-Altitude Intelligent Networking). *(Corresponding author: Guodong Jin.)*

Hanqing Zhang is with the Key Laboratory of Radar Imaging and Microwave Photonics, Ministry of Education, College of Electronic and Information Engineering, Nanjing University of Aeronautics and Astronautics, Nanjing 211106, China, and also with the AVIC Leihua Electronic Technology Research Institute, Wuxi 214083, China.

Guodong Jin and Daiyin Zhu are with the Key Laboratory of Radar Imaging and Microwave Photonics, Ministry of Education, College of Electronic and Information Engineering, Nanjing University of Aeronautics and Astronautics, Nanjing 211106, China, and also with Nanjing University of Aeronautics and Astronautics Shenzhen Research Institute, Shenzhen 518110, China (e-mail: jinguodong@nuaa.edu.cn).

Hongbo Zhang, Jie Lv, Yuan Cheng, Jing Ding, Zhenyu Guo, and Shaohua Ye are with the AVIC Leihua Electronic Technology Research Institute, Wuxi 214083, China.

Shilin Niu is with China United Network Communications Group Company Ltd., Henan Branch, Zhengzhou 450000, China.

Digital Object Identifier 10.1109/TGRS.2026.3692671

I. Introduction

SYNTHETIC aperture radar (SAR) is an advanced radar system that utilizes active microwave remote sensing technology to acquire target information [1], [2]. As a high-resolution imaging radar, SAR has all-weather detection capabilities and is not restricted by meteorological conditions. Therefore, SAR has become the core equipment of the global military reconnaissance system, playing an irreplaceable role in strategic intelligence acquisition and battlefield space control [3], [4], [5]. Currently, major countries around the world are actively promoting the research and engineering deployment of SAR technology to build strategic-level remote sensing capabilities for the future.

With the comprehensive upgrade of information-based electronic warfare, the operating frequency band of SAR systems faces severe challenges from a variety of complex electromagnetic threats. Without effective jamming suppression measures, SAR systems will struggle to acquire target information reliably. Therefore, improving the anti-jamming capability and practical performance of SAR in complex electromagnetic environments has important engineering value [6], [7].

A. Background

Digital radio frequency memory (DRFM) has become a critical technology in electronic warfare, and its integration with a jammer is advancing rapidly. The traditional 2-D interrupted sampling repeater jamming (2D-ISRJ) is an emerging form of intrapulse-modulated active coherent jamming enabled by DRFM technology [8]. The fundamental mechanism of 2D-ISRJ lies in the repeated sampling and forwarding of radar pulse signals. By using DRFM equipment, 2D-ISRJ achieves real-time sampling and forwarding processing of the intercepted radar transmit signal [9]. This operational mode significantly enhances jamming response speed and processing efficiency. Furthermore, by exploiting matched filtering characteristics, 2D-ISRJ generates multiple realistic

1558-0644 © 2026 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 02:12:53 UTC from IEEE Xplore. Restrictions apply.

false targets to deceive radar systems, while simultaneously bringing substantial challenges to the signal processing of the radar receiver [10].

Building on 2D-ISRJ, scholars worldwide are actively advancing research into composite jamming techniques by fusing the advantages of other jamming modalities to enhance jamming effectiveness. By dynamically tuning sampling-forwarding strategies of the jammer and optimizing sampled signal modulation parameters to generate agile and efficient composite jamming effects. Particularly for 2D-ISRJ based on noise multiplication modulation (NMM) [11], the combination of NMM with 2D-ISRJ effectively overcomes the inherent limitations of traditional 2D-ISRJ, such as distinct signal distribution patterns and obvious recognition features. Concurrently, its offline noise template enables dynamic and efficient generation of both false targets deception and dense false targets masking effects, while offering the dual advantages of flexible jamming forms and computationally efficient modulation processes.

B. Previous Work

To counter the rapid development of ISRJ in parameter optimization, complex modulation and multidimensional coordination, scholars worldwide predominantly focus on two key domains: waveform design at the transmitter and signal processing at the receiver. Based on the modulation characteristics of ISRJ, researchers have designed various waveform agility to suppress jamming [12], [13], [14], [15], [16]. Zhou et al. [12] proposed a novel approach employing a low-Doppler-tolerance phase-coded (PC) transmit waveform to suppress ISRJ, which significantly suppresses the processing gain of nonzero Doppler jamming components. Through the co-optimized design of PC waveform and mismatched filtering, this methodology effectively maintains high-quality imaging. To further enhance the real-time anti-ISRJ performance of the radar systems, Zhou et al. [13] established a co-optimization model integrating waveform agility and mismatched filtering, and combined the majorization minimization framework and squared iterative fast algorithm to achieve an efficient solution. Shi et al. [14] proposed an intelligent-driven notch linear frequency modulation (LFM) waveform design framework that estimates ISRJ parameters by the YOLOv7-tiny network. The notch waveform structure is reconstructed based on parameters, and time-shift extension processing is introduced to enhance the anti-ISRJ robustness. Su et al. [15] proposed an anti-ISRJ method utilizing complementary signal sets, which achieves the separation of target and jamming echo by designing a PC waveform that satisfies the strict complementary condition. Zhang et al. [16] proposed an anti-ISRJ method for multi-input multi-output (MIMO) radar based on array element carrier frequency and random pulse phase coding, which makes ISRJ tilted in the transmit range-angular frequency spectrum to separate targets and jamming, and combines the oblique projection operator with CFAR detection to achieve suppression.

Time–frequency (TF) domain signal processing is currently the mainstream method for radar receivers to anti-ISRJ [17], [18], [19], [20], [21], [22]. These methods use the TF segmentation characteristics of ISRJ to extract the jamming component and achieve jamming suppression by reconstruction or designing a bandpass filter. Gong et al. [17] designed a specific bandpass filter to suppress ISRJ by mining the TF discontinuity characteristics of the jamming signal. Wei et al. [18] realize ISRJ suppression based on a SAR system by designing a transmission waveform with variable subpulse time domain distribution and constructing an observation matrix to separate the undisturbed subpulse echo. Meng et al. [19] use a long-window short-time Fourier transform (STFT) to separate the TF distribution of the targets and ISRJ. It automatically generates the target TF mask by TFSegNet and reconstructs the desired signal using the inverse transform. Lu and Gao [20] proposed a method based on pseudorandom code continuous wave (PRC-CW) radar to achieve precise suppression by utilizing the peak fluctuation characteristics of sliding window pulse compression and combining it with the minimum residual criterion to finely reconstruct the ISRJ signal. Gao et al. [21] construct a binary TF template of the transmitted signal based on the frequency agile radar and combine the mean-variance dynamic criterion to accurately locate the target TF region. This method uses the inverse STFT to reconstruct the target signal and uses residual energy detection and range-dimensional peak filtering to achieve ISRJ residual depth suppression. In addition, Chen et al. [22] achieve ISRJ suppression in the spatial domain by multichannel array adaptive beamforming technology.

When composite jamming appears within the radar mainlobe beam, it exhibits significant overlap with the desired target in the TF domain and forms dense false targets in the spatial domain, rendering conventional anti-jamming methods ineffective [23], [24], [25]. Frequency diverse array MIMO (FDA-MIMO) SAR system [26], [27], [28], [29], [30], [31], [32], [33], [34] offers the potential to distinguish targets from mainlobe jamming in the transmit spatial domain, creating the possibility of suppressing mainlobe jamming in the spatial domain. The concept of the FDA was proposed in 2006 by Antonik et al. [35], which increased the controllable degrees of freedom (DOFs) in the range dimension by introducing a frequency step increment between array elements. Combining the FDA with MIMO technology not only enables effective separation of the transmission waveforms from each channel at the receiver [36], [37], [38], but also obtains the transmit steering vector with range-angle coupling characteristics, thereby significantly enhancing the flexibility of the radar system signal processing. However, research on composite jamming suppression methods based on FDA-MIMO SAR remains scarce [39], [40], [41], [42]. Furthermore, the traditional spatial domain composite jamming suppression methods suffer from several limitations, including the difficulty in obtaining a pure jamming-plus-noise covariance matrix, significant desired signal energy loss, and deterioration in beamforming performance [43].

To address the research gap in countering composite jamming within the FDA-MIMO SAR system and overcome the deficiencies of traditional spatial domain anti-jamming methods, this article proposes a mainlobe NMM-ISRJ suppression

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 02:12:53 UTC from IEEE Xplore. Restrictions apply.

method utilizing multistage blocking matrix preprocessing combined with covariance matrix symmetric extension (MBMP-CMSE) [44], [45]. This method first utilizes the MBMP to effectively suppress the desired signal while preserving the spatial characteristics of jamming and noise. Second, the CMSE technique is applied to significantly reduce the desired signal loss, thereby enhancing the anti-jamming performance of the proposed algorithm. Finally, the robustness of the beam-former is significantly improved by combining the norm constrained minimum variance distortionless response (NC-MVDR) beamforming algorithm [46].

C. Contribution and Organization

This article conducts in-depth research on the severe challenges faced by the FDA-MIMO SAR system in suppressing the mainlobe NMM-ISRJ composite jamming, and proposes an innovative joint processing framework. The main contributions and innovations of this work can be summarized as follows.

1) This article innovatively proposes the MBMP-CMSE method. The proposed method can nearly completely suppress the desired signal even when there is a direction of arrival (DOA) estimation error, while maximizing the retention of the spatial structure characteristics of the jamming and noise. Subsequently, through the CMSE technique, the reduced-dimensional covariance matrix is expanded back to the original array aperture dimension, effectively compensating for the array DOFs loss due to the multistage blocking operation, laying a solid foundation for the subsequent beamforming.

2) This article proposes to utilize the NC-MVDR algorithm to address issues such as main lobe offset and sidelobe rise. The NC-MVDR algorithm adds a quadratic inequality constraint (QIC) condition within the framework of the standard MVDR to limit the norm of the weighted vector. This constraint condition effectively enhances the fault tolerance of the beamformer for nonideal factors such as steering vector errors.

The organizational structure of this article's remaining sections is as follows. In Section II, the signal model of the FDA-MIMO SAR system is established, and the generation mechanism and jamming principle of NMM-ISRJ are analyzed, providing a theoretical basis for the subsequent anti-jamming algorithm. Section III proposes an anti-jamming method joint MBMP-CMSE and NC-MVDR. The principle of MBMP-CMSE is elaborated in detail, and NC-MVDR is introduced to improve the system's robustness. Section IV verifies the effectiveness of the proposed method in suppressing 2D-ISRJ and NMM-ISRJ through point-like targets and distributed scene simulation experiments. Quantitative analyses are carried out in terms of beamforming pattern, output signal-to-interference-plus-noise ratio (SINR), and imaging quality. Finally, Section V summarizes the main research findings of this article and emphasizes the advantages of the proposed method in suppressing the mainlobe composite jamming in FDA-MIMO SAR.

Notations: Boldface uppercase and lowercase letters are matrices and vectors, respectively; $(\cdot)^*$, $(\cdot)^T$, and $(\cdot)^H$ denote the conjugate, the transpose and the Hermitian transpose operations, respectively; $\odot$ and $\otimes$ represent Hadamard product and Kronecker product, respectively; $[\mathbf{a}]_n$ denotes the nth element of the vector $\mathbf{a}$; $[\mathbf{A}]_{n,m}$ denotes the nth row and mth column element of matrix $\mathbf{A}$; and $E[\cdot]$ represents mathematical expectation operator.

II. Signal Model and Jamming Principle of FDA-MIMO SAR

This section establishes the target mathematical signal model and analyzes the jamming mechanism of NMM-ISRJ in FDA-MIMO SAR. These analyses provide the theoretical support for the proposed mainlobe composite jamming suppression algorithm.

A. Signal Model of FDA-MIMO SAR

Without loss of generality, this section considers an FDA-MIMO SAR system with uniformly spaced linear arrays and co-located transmitters and receivers. The system consists of M transmitting elements and N receiving elements, with the interval between adjacent elements being d_a. Within the FDA array structure, the transmitting frequency of each transmitter increases linearly with a fixed frequency increment Δf [47]. Denoting the reference element (the first transmitter) with the carrier frequency f_0, the transmitting frequency of the mth element can be expressed as

$$f_m = f_0 + (m-1)\Delta f, \quad m = 1, \ldots, M \tag{1}$$

where $\Delta f \ll f_0$. The transmission signal of the mth transmitting unit can be expressed as

$$s_m(t_r) = \mathrm{rect}(t_r/T_r)\, x_m(t_r)\ \exp(j2\pi f_m t_r) \tag{2}$$

where t_r is the range fast time, rect$(\cdot)$ is the rectangular function, T_r is the pulse duration, and $x_m(t_r)$ denotes the transmission baseband waveforms. To ensure that the radar receivers effectively achieve the echo separation by using MIMO technology [48], the baseband waveforms transmitted by each transmitter should meet the following requirement, i.e.,

$$\int_{T_r} x_p(t)\, x_q^*(t-\tau)\, \mathrm{d}t = 0, \quad \text{when } p \neq q \tag{3}$$

where τ is the arbitrary time delay and $(\cdot)^*$ represents the conjugate operation. It should be noted that this article focuses on anti-jamming. Therefore, the baseband waveforms are presumed to satisfy the ideal orthogonality condition [49].

Fig. 1 shows the 3-D geometric configuration of the FDA-MIMO SAR system. To maintain generality, it is assumed that the FDA-MIMO SAR system is in the side-looking mode. SAR platform flies along the x-axis at speed v_r. y-axis points to the range and z-axis is perpendicular to the ground. The jammer is located at (x_j, y_j, z_j). The echo signal model is established at point $T(x_t, y_t, z_t)$. The received signal of the mth transmitter and nth receiver is given by [26], [27]

$$\begin{aligned} s_{mn}(t_r, t_a) = {} & \sigma_t \omega_{r,m}(t_r - \tau_{mn}(t_a))\, \omega_a(t_a)\, x_m(t_r - \tau_{mn}(t_a)) \\ & \times \exp\left[j2\pi (f_0 + (m-1)\Delta f)(t_r - \tau_{mn}(t_a))\right] \end{aligned} \tag{4}$$

where t_a is the azimuth slow time and σ_t denotes the scattering coefficient of the target. $\omega_{r,m}(t_r - \tau_{mn}(t_a))$ and $\omega_a(t_a)$ are the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 02:12:53 UTC from IEEE Xplore. Restrictions apply.

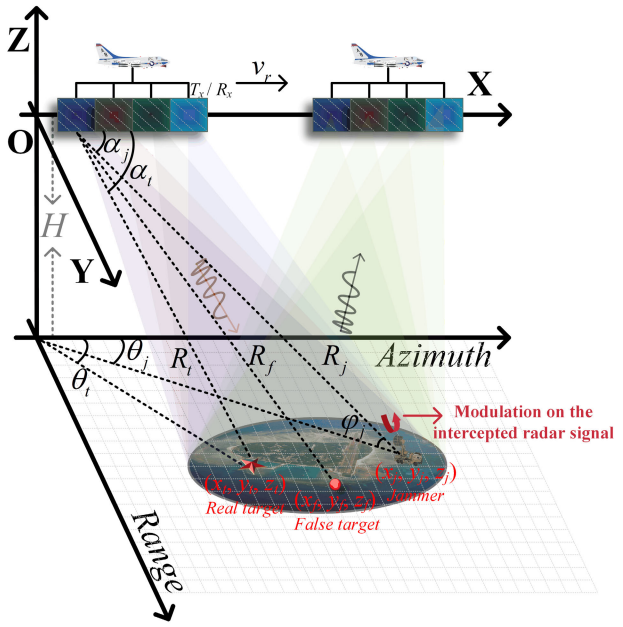


Fig. 1. 3-D geometry between the FDA-MIMO SAR system, a ground static scatterer, and a jammer.

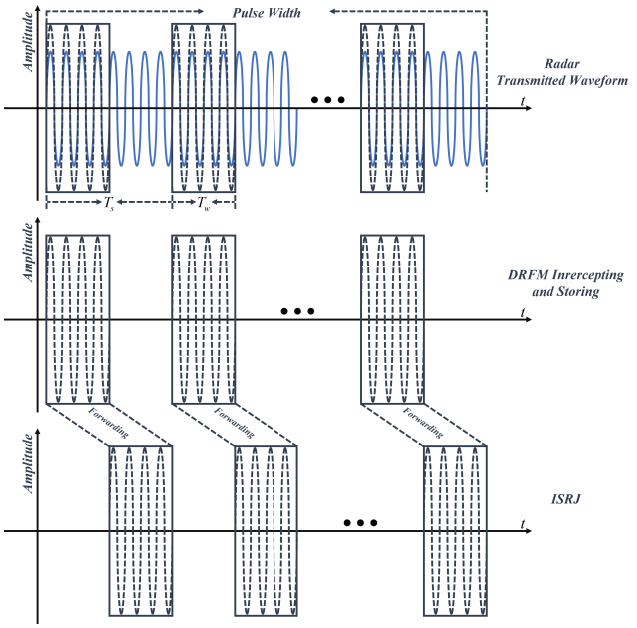


Fig. 2. Schematic of the working principle of interrupted sampling direct forwarding repeater jamming.

range and azimuth envelope, respectively. $\tau_{mn}(t_a)$ represents the time delay of the target echo, i.e.,

$$\tau_{mn}(t_a) = \frac{2R_t(t_a)}{c} - \frac{d_a(m-1)\cos(\alpha_t(t_a))}{c} - \frac{d_a(n-1)\cos(\alpha_t(t_a))}{c} \tag{5}$$

where c is the light speed, $\tau_t(t_a) = 2R_t(t_a)/c$ and $R_t(t_a)$ represents the slant range between the target and the reference element. $\alpha_t(t_a)$ denotes the conical angle between the target and reference element, i.e.,

$$\cos(\alpha_t(t_a)) = \cos(\theta_t(t_a))\cos(\psi_t(t_a)) \tag{6}$$

where $\theta_t(t_a)$ and $\psi_t(t_a)$ represent the azimuth and range angle of the target, respectively. [26] innovatively proposed the concept of the time index within pulse t', clarifying that the actual time t_r is composed of two parts: the transmit delay τ and the time index within pulse t', which satisfy the relationship $t_r = \tau + t'$. Based on this relationship, (4) can be rewritten as

$$\begin{aligned} s_{mn}(t_r, t_a) &\approx \sigma_t \omega_{r,m}(t_r - \tau_t(t_a))\,\omega_a(t_a)\,x_m(t_r - \tau_t(t_a))\,e^{j\phi_m(t_t')} \\ &= \sigma_t \omega_{r,m}(t_t')\,\omega_a(t_a)\,x_m(t_t')\,e^{j\phi_m(t_t')} \end{aligned} \tag{7}$$

where $t_t' = t_r - \tau_t(t_a)$ and $\phi_m(t_t')$ can be expressed as

$$\begin{aligned} \phi_m(t_t') &= 2\pi \int_0^{t_t' + \frac{d_a(m-1)\cos(\alpha_t(t_a))}{c} + \frac{d_a(n-1)\cos(\alpha_t(t_a))}{c}} \\ &\quad (f_0 + (m-1)\Delta f)\,\mathrm{d}t_t' \\ &\approx 2\pi \left(f_0 t_t' + (m-1)\Delta f t_t' + f_0 \frac{d_a(m-1)\cos(\alpha_t(t_a))}{c} \right. \\ &\quad \left. + f_0 \frac{d_a(n-1)\cos(\alpha_t(t_a))}{c} \right). \end{aligned} \tag{8}$$

Since $\Delta f \ll f_0$, (8) can be approximated, that is, $(m-1)\Delta f(d_a(m-1)\cos(\alpha_t(t_a)))/c$ and $(m-1)\Delta f(d_a(n-1)\cos(\alpha_t(t_a)))/c$ in (8) can be ignored. After matched filtering, the output of the mth transmitter and nth receiver can be expressed as

$$\begin{aligned} &s_{tmn}(t_t', t_a) \\ &\approx \sigma_t \omega_{r,m}(t_t')\,\omega_a(t_a)\,\xi_t \\ &\quad \times e^{j2\pi\left(-f_0\frac{2R_t(t_a)}{c} - (m-1)\Delta f\frac{2R_t(t_a)}{c} + f_0\frac{d_a(m-1)\cos(\alpha_t(t_a))}{c} + f_0\frac{d_a(n-1)\cos(\alpha_t(t_a))}{c}\right)} \end{aligned} \tag{9}$$

where ξ_t is the complex coefficient after matched filtering.

To facilitate subsequent analysis, the echoes in (9) can be superimposed into a matrix [42], i.e.,

$$\begin{aligned} \mathbf{Y}_t &= [s_{11}, s_{21}, \ldots, s_{M1}, s_{12}, \ldots, s_{mn}, \ldots, s_{MN}]^T \\ &= \kappa_t \mathbf{a}_t(\alpha_t(t_a)) \otimes \mathbf{b}_t(R_t(t_a), \alpha_t(t_a)) \end{aligned} \tag{10}$$

where κ_t indicates the complex amplitude and $(\cdot)^T$ denotes the transpose operation. $\mathbf{a}_t(\alpha_t(t_a))$ and $\mathbf{b}_t(R_t(t_a), \alpha_t(t_a))$ are the receive and transmit spatial frequency steering vector of the desired target, respectively, i.e.,

$$\mathbf{a}_t(\alpha_t(t_a)) = \left[1, e^{j2\pi f_r}, \ldots, e^{j2\pi(N-1)f_r}\right]^T \tag{11}$$

$$\mathbf{b}_t(R_t(t_a), \alpha_t(t_a)) = \left[1, e^{j2\pi f_t}, \ldots, e^{j2\pi(M-1)f_t}\right]^T \tag{12}$$

where $f_r = d_a\cos(\alpha_t(t_a))/\lambda$ and $f_t = d_a\cos(\alpha_t(t_a))/\lambda - 2\Delta f R_t(t_a)/c$ denote the receive and transmit spatial frequency of the desired target, respectively. $\lambda = c/f_0$ represents the wavelength.

B. Jamming Mechanism of NMM-ISRJ

Fig. 2 illustrates the interrupted sampling direct forwarding repeater jamming working principle. 2D-ISRJ is generated by DRFM detecting, sampling, storing, modulating, and forwarding intercepted radar transmission signals. Upon detecting the rising edge of the transmission pulse, the DRFM instantly samples and stores the signal. Subsequently, DRFM modulates

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 02:12:53 UTC from IEEE Xplore. Restrictions apply.

the stored slice signals according to the jamming strategy and forward them to the radar receiver. DRFM operates in an alternating sequence of sampling and forwarding. This process iterates until the radar transmission pulse falling edge is detected, thereby generating coherent jamming against the FDA-MIMO SAR system. A jammer is conventionally deployed within the radar mainbeam. Consequently, the jamming signal enters the receiver mainbeam concurrently with the target echo.

First, as shown in Fig. 1, the jammer intercepts the transmission signals of FDA-MIMO SAR and performs down-conversion, i.e.,

$$\begin{aligned} & s_{in}(t_r, t_a) \\ &= \sum_{m=1}^{M} \left[s_m(t_r) * \delta\left(t_r - \tau_{jm}(t_a)\right) \right] \exp(-j2\pi f_0 t_r) \\ &= \sum_{m=1}^{M} \operatorname{rect}\left(\frac{t_r - \tau_{jm}(t_a)}{T_r} \right) x_m\left(t_r - \tau_{jm}(t_a)\right) \\ & \quad \exp\left[-j2\pi f_0 \tau_{jm}(t_a)\right] \exp\left[j2\pi (m-1) \Delta f \left(t_r - \tau_{jm}(t_a)\right) \right] \end{aligned} \tag{13}$$

where the symbol "*" represents the convolution operation, $\delta(\cdot)$ is the impulse response function, and $\tau_{jm}(t_a)$ represents the time delay of the one-way slant range between the radar transmitters and the jammer, i.e.,

$$\tau_{jm}(t_a) = \frac{R_j(t_a)}{c} - \frac{d_a(m-1)\cos\left(\alpha_j(t_a)\right)}{c} \tag{14}$$

where $R_j(t_a)$ represents the slant range between the jammer and reference element.

Assuming that the ISRJ signal with a rectangular envelope can be modeled as a pulse train, its mathematical expression is

$$p(t) = \operatorname{rect}\left(\frac{t}{T_w} \right) * \sum_{i=-\infty}^{+\infty} \delta(t - iT_s) \tag{15}$$

where T_w and T_s denote the interrupted sampling pulsewidth and the interrupted sampling repetition period, respectively. By performing Fourier transform on (15), the spectrum of $p(t)$ can be obtained as follows:

$$\begin{aligned} P(f) &= \sum_{i=-\infty}^{+\infty} T_w f_s Sa(i\pi f_s T_w) \delta\left(f - if_s\right) \\ &= \sum_{i=-\infty}^{+\infty} a_n \delta\left(f - if_s\right) \end{aligned} \tag{16}$$

where $f_s = 1/T_s$ represents the interrupted sampling frequency and $a_n = T_w\ f_s\ Sa(i\pi f_s T_w)$ indicates the amplitude weighting coefficient. From this, it can be seen that interrupted sampling will generate a series of false target signals on the spectrum, which are symmetrically distributed on both sides of the desired target with sampling frequency f_s as the interval.

Second, DRFM interruptedly samples the intercepted radar transmission signals. Under the premise of ensuring the antenna transmission and reception isolation of the jammer, the impact of the range and azimuth ISRJ is independent to the fast and slow time of the echo phase history, respectively. That is, the 2D-ISRJ is equivalent to the series connection of the range and azimuth ISRJ. From this, it can be obtained that the 2D-ISRJ signal formed by DRFM is

$$s_{isrj}(t_r, t_a) = p_r(t_1) \cdot p_a(t_2) \cdot s_{in}(t_r, t_a) \tag{17}$$

where $p_r(t_1)$ and $p_a(t_2)$ are the range and azimuth interrupted sampling pulse train signal, respectively, i.e.,

$$p_r(t_1) = \operatorname{rect}\left(\frac{t_1}{T_{w1}} \right) * \sum_{z=-\infty}^{+\infty} \delta(t_1 - zT_{s1}) \tag{18}$$

$$p_a(t_2) = \operatorname{rect}\left(\frac{t_2}{T_{w2}} \right) * \sum_{k=-\infty}^{+\infty} \delta(t_2 - kT_{s2}). \tag{19}$$

After being processed by 2-D matched filtering, the 2D-ISRJ signal simultaneously forms net-like distribution jamming target groups in both range time and slow time, which has the effect of deception and suppression jamming.

Then, the jammer multiplies the 2D-ISRJ signal with the pregenerated noise modulation template $n(t_r, t_a)$, i.e.,

$$s_{nmm\text{-}isrj}(t_r, t_a) = n(t_r, t_a)\, s_{isrj}(t_r, t_a) \tag{20}$$

where $n(t_r, t_a)$ is uncorrelated in both fast and slow time. The jammer can perform 2-D filtering on the noise modulation template offline to control the jamming range. Subsequently, the filtered data is subjected to frequency domain sampling processing to generate false targets at designated locations. The generation process of the noise modulation template is shown in Appendix A and the error analysis is shown in Appendix B. By combining 2D-ISRJ with the NMM technology, NMM-ISRJ effectively overcomes the shortcomings of traditional 2D-ISRJ, such as the strong regularity and easy identification of the false targets. Furthermore, the jamming forms of NMM-ISRJ are flexible and varied, and the modulation process is simple and efficient.

Finally, the jammer up-converts the NMM-ISRJ signal and forward it to the FDA-MIMO SAR system, i.e.,

$$s_j(t_r, t_a) = \left[s_{nmm\text{-}isrj}(t_r, t_a) * \delta\left(t_r - \Delta T_j(t_a)\right) \right] \exp(j2\pi f_0 t_r) \tag{21}$$

where $\Delta T_j(t_a) = 2[R_f(t_a) - R_j(t_a)]/c$ is the jammer modulated time delay. $R_f(t_a)$ represents the slant range between the false targets and the reference element. The mathematical model of the NMM-ISRJ echo received by the FDA-MIMO SAR system is

$$s_{jn}(t_r, t_a) = \left[s_j(t_r, t_a) * \delta\left(t_r - \tau_{jn}(t_a)\right) \right] \times \exp(-j2\pi f_0 t_r) \tag{22}$$

where $\tau_{jn}(t_a)$ denotes the time delay of the jammer forwarding to the nth receiver, i.e.,

$$\tau_{jn}(t_a) = \frac{R_j(t_a)}{c} - \frac{d_a(n-1)\cos\left(\alpha_t(t_a)\right)}{c}. \tag{23}$$

Consistent with the analysis in Section II-A, after matched filtering, the output of the mth transmitter and nth receiver is given by [26]

$$s_{jmn}\left(t_f^{'}, t_a\right) \approx \sigma_f \omega_{r,m}\left(t_f^{'}\right) \omega_a(t_a)\, \xi_f$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 02:12:53 UTC from IEEE Xplore. Restrictions apply.

$$\times e^{j2\pi\left(-f_0\frac{2R_j(t_a)}{c}-(m-1)\Delta f\frac{2R_f(t_a)}{c}+f_0\frac{d_a(m-1)\cos\left(\alpha_j(t_a)\right)}{c}+f_0\frac{d_a(n-1)\cos\left(\alpha_j(t_a)\right)}{c}\right)} \tag{24}$$

where

$$\begin{aligned}\tau_{jmn}(t_a) &= \tau_{jm}(t_a)+\tau_{jn}(t_a)+\Delta T_j(t_a)\\ &= \frac{2R_f(t_a)}{c}-\frac{d_a(m-1)\cos(\alpha_j(t_a))}{c}\\ &\quad -\frac{d_a(n-1)\cos(\alpha_j(t_a))}{c}\end{aligned} \tag{25}$$

where $t_f^{'}=t_r-\tau_f(t_a)$ and $\tau_f(t_a)=2R_f(t_a)/c$. $R_f(t_a)$ represents the slant range between the false targets and the reference element.

Similar to formula (10), (24) can be rearranged as

$$\begin{aligned}\mathbf{Y}_j &= \left[s_{j11},s_{j21},\ldots,s_{jM1},s_{j12},\ldots,s_{jmn},\ldots,s_{jMN}\right]^T\\ &= \kappa_f\mathbf{a}_f\left(\alpha_j(t_a)\right)\otimes\mathbf{b}_f\left(R_f(t_a),\alpha_j(t_a)\right)\end{aligned} \tag{26}$$

where $\mathbf{a}_f(\alpha_j(t_a))$ and $\mathbf{b}_f(R_f(t_a),\alpha_j(t_a))$ can be, respectively, expressed as

$$\mathbf{a}_f\left(\alpha_j(t_a)\right)=\left[1,e^{j2\pi f_{jr}},\ldots,e^{j2\pi(N-1)f_{jr}}\right]^T \tag{27}$$

$$\mathbf{b}_f\left(R_f(t_a),\alpha_j(t_a)\right)=\left[1,e^{j2\pi f_{jt}},\ldots,e^{j2\pi(M-1)f_{jt}}\right]^T \tag{28}$$

where $f_{jr}=d_a\cos(\alpha_j(t_a))/\lambda$ and $f_{jt}=d_a\cos(\alpha_j(t_a))/\lambda-2\Delta fR_f(t_a)/c$ denote the receive and transmit spatial frequency of the false targets, respectively.

It is worth noting that the jamming signals generated by the same jammer have the same receiving spatial frequency. By comparing f_r and f_{jr}, it can be seen that the two are highly similar in mathematical form. In fact, in order to separate the desired target from the composite jamming, a range compensation strategy can be adopted to effectively distinguish the desired signal from NMM-ISRJ, thereby reducing the complexity of subsequent signal processing. f_t in (12) compensated by the range cell can be expressed as

$$f_t=d_a\cos\left(\alpha_t(t_a)\right)/\lambda. \tag{29}$$

The jammer uses modulation techniques to alter the range and Doppler information of the intercepted signal from its own position to the false target's position [42]. However, (24) indicates that the spatial information of the jammer is still retained in the jamming echo. When range compensation is carried out using the phase corresponding to the jammer position, the transmit spatial frequency of the jamming signal will be compensated as

$$f_{jt}=d_a\cos\left(\alpha_j(t_a)\right)/\lambda-2\Delta f\left(R_f(t_a)-R_j(t_a)\right)/c. \tag{30}$$

As indicated by (30), the time-spatial information mismatch of NMM-ISRJ introduces a residual phase component after range compensation.

After transmit spatial frequency compensation, the transmit–receive spatial frequency of the desired target is completely consistent, showing a diagonal distribution in Fig. 3. On the contrary, the jamming signal exhibits a discrete distribution in Fig. 3 due to the presence of range compensation margin in the transmit spatial frequency [42].

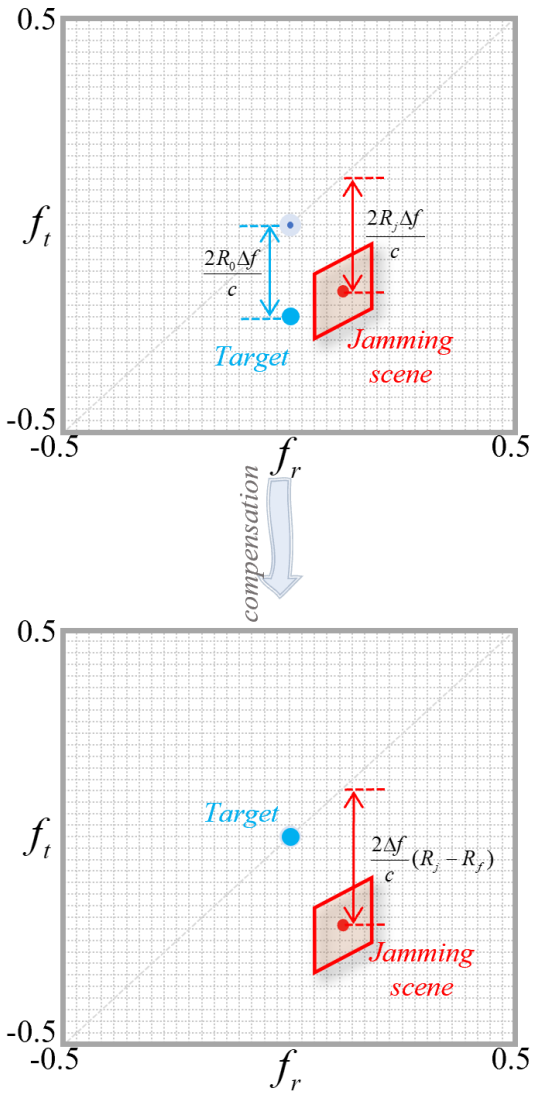


Fig. 3. Desired target and jamming distribution in 2-D transmit–receive spatial frequency before and after range compensation.

For notational simplicity, the mixed signal received by FDA-MIMO SAR can be expressed as

$$\mathbf{x}=\mathbf{A}_N\mathbf{s}_{tf}+\mathbf{n} \tag{31}$$

where $\mathbf{A}_N=[\mathbf{a}_{N,t},\mathbf{a}_{N,f}]$ denotes the steering vector matrix of receive spatial frequency. $\mathbf{a}_{N,t}=\mathbf{a}_t(\alpha_t(t_a))$ and $\mathbf{a}_{N,f}=\mathbf{a}_f(\alpha_j(t_a))$. $\mathbf{s}_{tf}=[\mathbf{s}_t,\mathbf{s}_f]^T$, $\mathbf{s}_t$ and $\mathbf{s}_f$ are the desired target signal and jamming signal, respectively. $\mathbf{n}$ represents the system noise matrix.

III. Proposed Anti-NMM-ISRJ Algorithm

Based on the analysis of the desired target and mainlobe NMM-ISRJ characteristics and generation mechanism in Section II. This section proposes the joint MBMP-CMSE and NC-MVDR method to achieve jamming suppression in f_t-f_r domain.

A. Principle of Joint MBMP and CMSE

The traditional BMP and IBMP preprocesses the received signal by designing the blocking matrix $\mathbf{B}_{bmp}$ and $\mathbf{B}_{ibmp}$,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 02:12:53 UTC from IEEE Xplore. Restrictions apply.

respectively. The construction of $\mathbf{B}_{bmp}$ and $\mathbf{B}_{ibmp}$ can be expressed as

$$\mathbf{B}_{bmp} = \begin{bmatrix} 1 & -e^{-j\rho_1} & 0 & \cdots & 0 \\ 0 & 1 & -e^{-j2\rho_1} & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 & -e^{-j\rho_1} \end{bmatrix}_{(N-1)\times N} \tag{32}$$

$$\mathbf{B}_{ibmp} = \begin{bmatrix} 1 & 0 & \cdots & 0 & -e^{-j\frac{N}{2}\rho_1} & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 & 0 & -e^{-j\frac{N}{2}\rho_1} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 & 0 & 0 & \cdots & -e^{-j\frac{N}{2}\rho_1} \end{bmatrix}_{\frac{N}{2}\times N} \tag{33}$$

where $\rho_1 = 2\pi(d_a \cos(\tilde{\alpha}_t))/\lambda$ and $\tilde{\alpha}_t(t_a)$[1] denotes the estimated DOA value of the desired target. The principle of this blocking matrix is to block the desired signal by using the cancellation processing of adjacent antenna elements. However, when $\tilde{\alpha}_t(t_a)$ has estimation error, the traditional BMP and IBMP cannot completely eliminate the desired signal. This leads to its leakage into the jamming-plus-noise subspace, where it cancels out the coherent jamming, resulting in a degradation of beamforming performance. Meanwhile, as can be seen from (32) and (33), the data processed by traditional BMP and IBMP will lose one and $N/2$ antenna DOFs, respectively. The array DOFs loss can lead to issues such as mainlobe offset and sidelobe level rise of the processed data.

To address the above issues, this section first proposes the MBMP method to fully eliminate the desired signal. On this basis, CMSE is further proposed to overcome the mainlobe offset and sidelobe level rise caused by MBMP. To completely eliminate the desired signal, the multistage blocking matrix structure $\mathbf{B}_1, \ldots, \mathbf{B}_{N/2}$ are used to preprocess the received signal. The general expression can be defined as

$$\mathbf{B}_k = \begin{bmatrix} 1 & -e^{-j\rho_1} & 0 & \cdots & 0 \\ 0 & 1 & -e^{-j\rho_1} & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 & -e^{-j2\rho_1} \end{bmatrix}_{(N-k)\times(N-k+1)} \tag{34}$$

where $k = 1, \ldots, N/2$.

Assuming that the output signal $\mathbf{x}_k$ of the kth stage is blocking. The dimension of $\mathbf{x}_k$ is reduced to $(N-k)\times 1$ after blocking operation. At the same time, $\mathbf{x}_k$ is also the input signal for the $(k+1)$th stage blocking. To maintain generality, we will now use the kth stage of blocking as an example to illustrate the principle of multilevel blocking. In order to maintain universality, the principle of MBMP can be explained by taking the kth stage blocking as an example. According to the previous analysis, $\mathbf{x}_{k-1}$ is the input signal for the kth stage blocking. Taking the first $N-k$ and the last $N-k$ elements of $\mathbf{x}_{k-1}$ in sequence to form $\mathbf{x}_{k-1}^1$ and $\mathbf{x}_{k-1}^2$, respectively. Then, $\mathbf{x}_k$ can be calculated as

$$\mathbf{x}_k = \mathbf{x}_{k-1}^1 - e^{-j\rho_1}\mathbf{x}_{k-1}^2. \tag{35}$$

Equation (35) can be rewritten in the form of the blocking matrix operation and further expressed as

$$\mathbf{x}_k = \mathbf{B}_k\mathbf{x}_{k-1}. \tag{36}$$

[1]The conical angle $\tilde{\alpha}_t(t_a)$ of target is a function of the slow time t_a within the synthetic aperture time.

From the above analysis, it can be inferred that the continuous $N/2$ stage blocking is the continuous multiplication of $N/2$ blocking matrix. Therefore, (36) can be recalculated as

$$\begin{aligned} \mathbf{x}_{N/2} &= \mathbf{B}_{N-N/2}\mathbf{B}_{N/2-1}, \ldots, \mathbf{B}_1\mathbf{x} \\ &= \mathbf{B}_{N/2}\mathbf{x} \end{aligned} \tag{37}$$

where

$$\mathbf{B}_{N/2} = \mathbf{B}_{N-N/2}\mathbf{B}_{N/2-1}, \ldots, \mathbf{B}_1. \tag{38}$$

If the influence of radar system noise is ignored, the output of the first stage blocking is

$$\begin{aligned} \mathbf{x}_1 &= \mathbf{B}_1\mathbf{x} = \mathbf{B}_1\left[\mathbf{a}_{N,t}\mathbf{s}_t + \mathbf{a}_{N,f}\mathbf{s}_f\right] \\ &= \left(1 - e^{-j2\pi d_a \cos(\tilde{\alpha}_t)/\lambda}\mu_1\right)\mathbf{a}_{N-1,t}\mathbf{s}_t \\ &\quad + \left(1 - e^{-j2\pi d_a \cos(\tilde{\alpha}_t)/\lambda}\mu_2\right)\mathbf{a}_{N-1,f}\mathbf{s}_f \\ &= \beta_1\mathbf{a}_{N-1,t}\mathbf{s}_t + \beta_2\mathbf{a}_{N-1,f}\mathbf{s}_f \end{aligned} \tag{39}$$

where $\mu_i = e^{j2\pi d_a \cos(\alpha_i)/\lambda}, i = 1, 2$, $\alpha_1 = \tilde{\alpha}_t + \Delta\alpha$ and $\alpha_2 = \alpha_j$ represents the true DOA value of the desired target and jamming signal, respectively. $\Delta\alpha$ is the DOA estimation error of the desired target signal. $\beta_i = 1 - e^{-j2\pi d_a \cos(\tilde{\alpha}_t)/\lambda}\mu_i, i = 1, 2$. (39) extends to the second stage blocking, i.e.,

$$\begin{aligned} \mathbf{x}_2 &= \mathbf{B}_2\mathbf{x}_1 = \mathbf{B}_2\left[\beta_1\mathbf{a}_{N-1,t}\mathbf{s}_t + \beta_2\mathbf{a}_{N-1,f}\mathbf{s}_f\right] \\ &= \left(1 - e^{-j2\pi d_a \cos(\tilde{\alpha}_t)/\lambda}\mu_1\right)\beta_1\mathbf{a}_{N-2,t}\mathbf{s}_t \\ &\quad + \left(1 - e^{-j2\pi d_a \cos(\tilde{\alpha}_t)/\lambda}\mu_2\right)\beta_2\mathbf{a}_{N-2,f}\mathbf{s}_f \\ &= \beta_1^2\mathbf{a}_{N-2,t}\mathbf{s}_t + \beta_2^2\mathbf{a}_{N-2,f}\mathbf{s}_f. \end{aligned} \tag{40}$$

Similarly, the output signal of the $N/2$ stage blocking can be calculated as

$$\mathbf{x}_{N/2} = \beta_1^{\frac{N}{2}}\mathbf{a}_{\frac{N}{2},t}\mathbf{s}_t + \beta_2^{\frac{N}{2}}\mathbf{a}_{\frac{N}{2},f}\mathbf{s}_f. \tag{41}$$

Based on the above analysis, β_1 and β_2 can be obtained, respectively,

$$\begin{aligned} \beta_1 &= 1 - e^{-j2\pi d_a \cos(\tilde{\alpha}_t)/\lambda}e^{j2\pi d_a \cos(\alpha_t+\Delta\alpha)/\lambda} \\ &= 1 - e^{j2\pi d_a/\lambda[\cos(\alpha_t+\Delta\alpha)-\cos(\tilde{\alpha}_t)]} \\ \beta_2 &= 1 - e^{-j2\pi d_a \cos(\tilde{\alpha}_t)/\lambda}e^{j2\pi d_a \cos(\alpha_2)/\lambda} \\ &= 1 - e^{j2\pi d_a/\lambda[\cos(\alpha_2)-\cos(\tilde{\alpha}_t)]}. \end{aligned} \tag{42}$$

Due to $\Delta\alpha$ is small and $\cos(\alpha_t + \Delta\alpha) - \cos(\tilde{\alpha}_t) < \cos(\alpha_2) - \cos(\tilde{\alpha}_t)$, it can be inferred from (42) that

$$1 > |\beta_2| > |\beta_1|. \tag{43}$$

Equation (43) indicates that when the blocking level increases to $N/2$, β_1 quickly converges to zero, while the convergence of β_2 is relatively limited. After $N/2$ stage blocking processing, the desired signal in the output signal is effectively suppressed, mainly including the jamming-plus-noise signal.

In summary, the MBMP algorithm can suppress the desired signal while maintaining the spatial characteristics of jamming-plus-noise unchanged, significantly improving the robustness of the algorithm. However, the MBMP algorithm has a significant problem of array aperture DOFs loss: for each stage of blocking operation experienced by the echo signal, its dimension decreases by one dimension. After $N/2$ stage blocking operation, the dimension of $\mathbf{R}_{N/2}$ is reduced to $N/2 \times 1$, which can only perform weighted synthesis on the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 02:12:53 UTC from IEEE Xplore. Restrictions apply.

received signals of $N/2$ subarray, resulting in a loss of half the array aperture DOFs. To reduce the loss of array aperture DOFs, this section proposes a CMSE algorithm.

The covariance matrix of jamming-plus-noise signal processed by MBMP can be calculated as

$$\begin{aligned}\mathbf{R}_{N/2} &= \mathbf{E}\left[\mathbf{B}_{N/2}\mathbf{x}\mathbf{x}^H\mathbf{B}_{N/2}^H\right] \\ &= \mathbf{B}_{N/2}\mathbf{A}_N\mathbf{E}\left[\mathbf{s}_{tf}\mathbf{s}_{tf}^H\right]\mathbf{A}_N^H\mathbf{B}_{N/2}^H + \mathbf{E}\left[\mathbf{n}\mathbf{n}^H\right] \\ &= \mathbf{B}_{N/2}\mathbf{A}_N\mathbf{R}_s\mathbf{A}_N^H\mathbf{B}_{N/2}^H + 2\sigma_n^2\mathbf{I} \end{aligned} \tag{44}$$

where $E[\cdot]$ represents mathematical expectation operator, $(\cdot)^H$ denotes the Hermitian transpose operation and $\mathbf{R}_s$ represents the auto-correlation matrix of the signals. $\mathbf{I}$ is the unit matrix and its dimension is $N/2 \times N/2$. The covariance conjugate matrix of (44) can be expressed as

$$\begin{aligned}\mathbf{R}_{N/2}^* &= \left(\mathbf{B}_{N/2}\mathbf{A}_N\mathbf{R}_s\mathbf{A}_N^H\mathbf{B}_{N/2}^H\right)^* + \left(2\sigma_n^2\mathbf{I}\right)^* \\ &= \mathbf{B}_{N/2}^*\mathbf{A}_N^*\mathbf{R}_s^*\mathbf{A}_N^T\mathbf{B}_{N/2}^T + 2\sigma_n^2\mathbf{I}. \end{aligned} \tag{45}$$

Due to the signals received by the array are independent of each other, it can be assumed that $\mathbf{R}_s = \mathbf{R}_s^*$. Equation (45) can be rewritten as

$$\mathbf{R}_{N/2}^* = \mathbf{B}_{N/2}^*\mathbf{A}_N^*\mathbf{R}_s\mathbf{A}_N^T\mathbf{B}_{N/2}^T + 2\sigma_n^2\mathbf{I}. \tag{46}$$

Combining (44)–(46), the new data matrix can be expressed as

$$\begin{aligned}\mathbf{Y}^H &= \left[\mathbf{R}_{N/2}^*\mathbf{J}_{N/2}\mathbf{R}_{N/2}\right] \\ &= \left[\mathbf{B}_{N/2}^*\mathbf{A}_N^*\mathbf{R}_s\mathbf{A}_N^T\mathbf{B}_{N/2}^T\mathbf{J}_{N/2} + 2\sigma_n^2\mathbf{I}\mathbf{J}_{N/2}\right. \\ &\qquad \left.\mathbf{B}_{N/2}\mathbf{A}_N\mathbf{R}_s\mathbf{A}_N^H\mathbf{B}_{N/2}^H + 2\sigma_n^2\mathbf{I}\right] \\ &= \left[\mathbf{Z}^*\mathbf{R}_s\mathbf{Z}^T\mathbf{J}_{N/2}\mathbf{Z}\mathbf{R}_s\mathbf{Z}^H\right] + 2\sigma_n^2\left[\mathbf{I}\mathbf{J}_{N/2}\mathbf{I}\right] \end{aligned} \tag{47}$$

where $\mathbf{Z} = \mathbf{B}_{N/2}\mathbf{A}_N$, $\mathbf{J}_{N/2}$ denotes the reverse unit matrix and its dimension is $N/2 \times N/2$.

The extended covariance matrix of jamming-plus-noise can be expressed as

$$\begin{aligned}\mathbf{R}_{se} &= \mathbf{E}\left[\mathbf{Y}\mathbf{Y}^H\right] \\ &= \begin{bmatrix}\mathbf{J}_{\frac{N}{2}}\mathbf{Z}^*\mathbf{R}_s\mathbf{Z}^T\mathbf{Z}^*\mathbf{R}_s\mathbf{Z}^T\mathbf{J}_{\frac{N}{2}} & \mathbf{J}_{\frac{N}{2}}\mathbf{Z}^*\mathbf{R}_s\mathbf{Z}^T\mathbf{Z}\mathbf{R}_s\mathbf{Z}^H \\ \mathbf{Z}\mathbf{R}_s\mathbf{Z}^H\mathbf{Z}^*\mathbf{R}_s\mathbf{Z}^T\mathbf{J}_{\frac{N}{2}} & \mathbf{Z}\mathbf{R}_s\mathbf{Z}^H\mathbf{Z}\mathbf{R}_s\mathbf{Z}^H\end{bmatrix} \\ &\quad + 4\sigma_n^4\begin{bmatrix}\mathbf{J}_{\frac{N}{2}}\mathbf{I} \\ \mathbf{I}\end{bmatrix}\left[\mathbf{I}\mathbf{J}_{\frac{N}{2}}\mathbf{I}\right] \end{aligned} \tag{48}$$

where $\mathbf{Y} = [\mathbf{Y}^H]^H$.

For the convenience of analysis, (48) can be simplified as

$$\begin{aligned}\mathbf{R}_{se} &= \mathbf{A}_{se}\begin{bmatrix}\mathbf{B}_{\frac{N}{2}}^*\mathbf{R}_s\mathbf{Z}^T\mathbf{Z}^*\mathbf{R}_s\mathbf{B}_{\frac{N}{2}}^T & \mathbf{B}_{\frac{N}{2}}^*\mathbf{R}_s\mathbf{Z}^T\mathbf{Z}\mathbf{R}_s\mathbf{B}_{\frac{N}{2}}^H \\ \mathbf{B}_{\frac{N}{2}}\mathbf{R}_s\mathbf{Z}^H\mathbf{Z}^*\mathbf{R}_s\mathbf{B}_{\frac{N}{2}}^T & \mathbf{B}_{\frac{N}{2}}\mathbf{R}_s\mathbf{Z}^H\mathbf{Z}\mathbf{R}_s\mathbf{B}_{\frac{N}{2}}^H\end{bmatrix}\mathbf{A}_{se}^H \\ &\quad + 4\sigma_n^4\mathbf{N}_{se}\mathbf{N}_{se}^H \end{aligned} \tag{49}$$

where $\mathbf{A}_{se} = \begin{bmatrix}\mathbf{J}_{N/2}\mathbf{A}_N^* \\ \mathbf{A}_N\end{bmatrix}$ and $\mathbf{N}_{se} = \begin{bmatrix}\mathbf{J}_{N/2}\mathbf{I} \\ \mathbf{I}\end{bmatrix}$. Due to the CMSE algorithm will convert white noise into colored noise, it is necessary to whiten the expanded data [45], i.e.,

$$\mathbf{\Gamma}\left(\sigma_n^4\mathbf{N}_{se}\mathbf{N}_{se}^H\right)\mathbf{\Gamma}^H = \sigma_n^2\mathbf{I}. \tag{50}$$

B. Principle of NC-MVDR

Directly applying conventional adaptive beamforming to the output data of the MBMP-CMSE algorithm will result in problems such as main lobe shift and elevated sidelobe levels [50]. Therefore, this section introduces a robust NC-MVDR beamforming algorithm. This algorithm introduces a new QIC condition on the weighted vector under the standard MVDR framework, which can significantly improve the robustness of the beamforming to beam pointing errors and signal mismatched. The standard MVDR beamforming problem can be formulated as

$$\begin{cases}\min\ \mathbf{w}^H\mathbf{R}\mathbf{w} \\ \text{s.t. } \mathbf{w}^H\mathbf{a} = 1.\end{cases} \tag{51}$$

Then, the optimal weighted vector of the standard MVDR can be expressed as

$$\mathbf{w} = \frac{\mathbf{R}^{-1}\mathbf{a}}{\mathbf{a}^H\mathbf{R}^{-1}\mathbf{a}}. \tag{52}$$

By introducing a QIC condition, the NC-MVDR beamforming algorithm can be described as

$$\begin{cases}\min\ \mathbf{w}^H\mathbf{R}_{se}\mathbf{w} \\ \text{s.t. } \mathbf{w}^H\mathbf{a}_{N,t} = 1 \\ \|\ \mathbf{w}\ \|^2 \le \xi\end{cases} \tag{53}$$

where ξ denotes the constraint parameter. The Lagrange multiplier method can be used to transform (53) into solving the function extreme problem about $\mathbf{w}, \rho$ and μ, i.e.,

$$\begin{aligned}&g\left(\mathbf{w},\rho,\mu\right) \\ &= \mathbf{w}^H\mathbf{R}_{se}\mathbf{w} + \rho\left(\|\ \mathbf{w}\ \|^2 - \xi\right) + \mu\left(-\mathbf{w}^H\mathbf{a}_{N,t} - \mathbf{a}_{N,t}^H\mathbf{w} + 2\right) \end{aligned} \tag{54}$$

where ρ is the Lagrange multiplier, which meets $\rho \ge 0$ and $\mathbf{R} + \rho\mathbf{I} > 0$. μ is the Lagrange multiplier and μ is a arbitrary value.

Next, the process of minimizing the weighted vector $\mathbf{w}$ in (54) will be analyzed and derived. When $\rho = 0$, that is, $\rho(\|\ \mathbf{w}\ \|^2 - \xi) = 0$. The solution of NC-MVDR is equivalent to the solution of standard MVDR. The above QIC constraint $\|\ \mathbf{w}\ \|^2 \le \xi$ are invalid, which cannot effectively constrain the weight $\mathbf{w}$. This will reduce the robustness of the NC-MVDR algorithm.

When $\rho > 0$, in order to improve the robustness of the NC-MVDR algorithm, ξ should satisfy

$$\xi < \frac{\mathbf{a}_{N,t}^H\mathbf{R}_{se}^{-2}\mathbf{a}_{N,t}}{\left(\mathbf{a}_{N,t}^H\mathbf{R}_{se}^{-1}\mathbf{a}_{N,t}\right)^2}. \tag{55}$$

This condition constrains the upper limit of ρ. Under this constraint, (54) can be rewritten as

$$\begin{aligned}g\left(\mathbf{w},\rho,\mu\right) &= \left[\mathbf{w} - \mu\left(\mathbf{R}_{se} + \rho\mathbf{I}\right)^{-1}\mathbf{a}_{N,t}\right]^H\left(\mathbf{R}_{se} + \rho\mathbf{I}\right) \\ &\quad \left[\mathbf{w} - \mu\left(\mathbf{R}_{se} + \rho\mathbf{I}\right)^{-1}\mathbf{a}_{N,t}\right] \\ &\quad - \mu^2\mathbf{a}_{N,t}^H\left(\mathbf{R}_{se} + \rho\mathbf{I}\right)^{-1}\mathbf{a}_{N,t} - \rho\xi + 2\mu. \end{aligned} \tag{56}$$

It can be seen from (56) that for given ρ and μ, g takes the minimum value when $\hat{\mathbf{w}} = \mu(\mathbf{R}_{se} + \rho\mathbf{I})^{-1}\mathbf{a}_{N,t}$, i.e.,

$$g_1\left(\rho,\mu\right) = -\mu^2\mathbf{a}_{N,t}^H\left(\mathbf{R}_{se} + \rho\mathbf{I}\right)^{-1}\mathbf{a}_{N,t} - \rho\xi + 2\mu. \tag{57}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 02:12:53 UTC from IEEE Xplore. Restrictions apply.

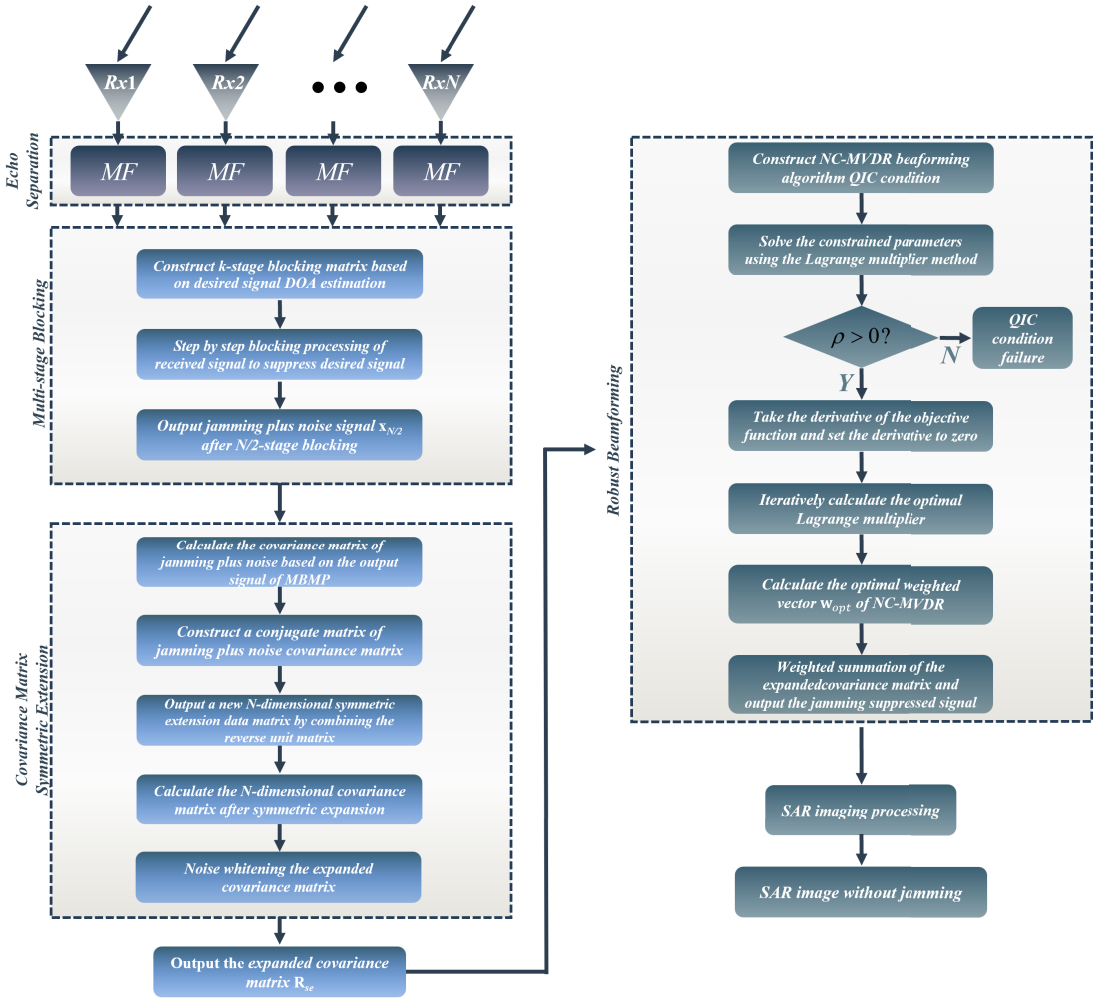


Fig. 4. Joint MBMP-CMSE and NC-MVDR technologies for suppressing the mainlobe NMM-ISRJ process diagram.

By differentiating (57) with respect to μ and setting the result to zero, μ can be calculated as

$$\hat{\mu} = \frac{1}{\mathbf{a}_{N,t}^H \left(\mathbf{R}_{se} + \rho \mathbf{I}\right)^{-1} \mathbf{a}_{N,t}}. \tag{58}$$

Substituting (58) into (57), i.e.,

$$g_2(\rho) = -\rho\xi + \frac{1}{\mathbf{a}_{N,t}^H \left(\mathbf{R}_{se} + \rho \mathbf{I}\right)^{-1} \mathbf{a}_{N,t}}. \tag{59}$$

By differentiating (59) with respect to ρ and setting the result to zero, ξ can be calculated as

$$\xi = \frac{\mathbf{a}_{N,t}^H \left(\mathbf{R}_{se} + \hat{\rho} \mathbf{I}\right)^{-2} \mathbf{a}_{N,t}}{\left[\mathbf{a}_{N,t}^H \left(\mathbf{R}_{se} + \hat{\rho} \mathbf{I}\right)^{-1} \mathbf{a}_{N,t}\right]^2} \tag{60}$$

where the detailed solution process for the value of $\hat{\rho}$ can be referred to in [46].

Substituting $\hat{\mu}$ into $\hat{\mathbf{w}} = \mu(\mathbf{R}_{se} + \rho\mathbf{I})^{-1}\mathbf{a}_{N,t}$, the optimal weighted vector can be calculated as

$$\mathbf{w}_{opt} = \frac{\left(\mathbf{R}_{se} + \hat{\rho} \mathbf{I}\right)^{-1} \mathbf{a}_{N,t}}{\mathbf{a}_{N,t}^H \left(\mathbf{R}_{se} + \hat{\rho} \mathbf{I}\right)^{-1} \mathbf{a}_{N,t}}. \tag{61}$$

After NC-MVDR processing, the jamming signal will be suppressed by the deep nulls, thus avoiding the influence on the desired signal. It should be noted that the scene information obtained at the location of the jammer has been preserved. Fig. 4 systematically illustrates the overall processing flow of the proposed joint MBMP-CMSE and NC-MVDR method. This flowchart gradually demonstrates the core steps of the proposed anti-jamming algorithm with a clear modular structure.

C. Computational Complexity

To comprehensively evaluate the computational efficiency of the proposed algorithm, this section will conduct a systematic analysis of its computational complexity, providing quantitative evidence to support the feasibility and applicability of the proposed algorithm in practical systems. First, the computational complexity of MBMP can be calculated as $O(L \cdot N^2/4)$, where L represents the snapshots. Second, the computational complexity of CMSE can be calculated in four parts, which are the covariance matrix of jamming-plus-noise signal $\mathbf{R}_{N/2}$ estimation, the new data matrix $\mathbf{Y}^H$ construction, the $\mathbf{Y}^H$ matrix transposition, and the extended covariance matrix of jamming-plus-noise signal $\mathbf{R}_{se}$ calculation, respectively. The computational complexity of these parts are $O(L^2 \cdot N^2/4)$, $O(N^2/4) + O(N^3/8)$, $O(N^2/4) + O(N^2/4)$, and $O(L \cdot N^3/8)$, respectively. Finally, the main computational requirement of NC-MVDR comes from the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 02:12:53 UTC from IEEE Xplore. Restrictions apply.

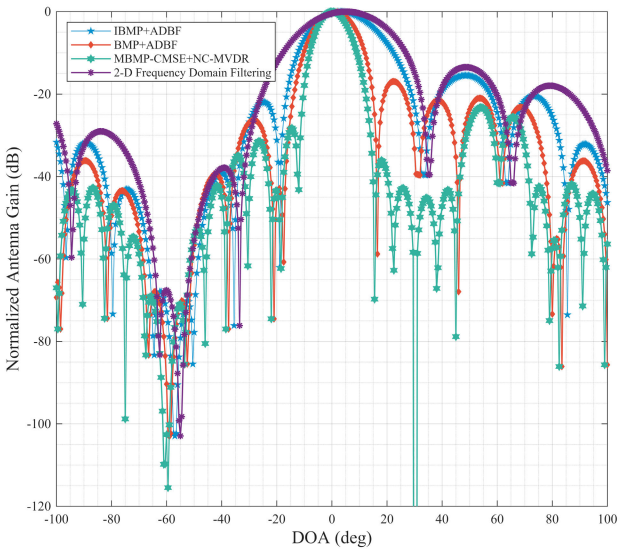


Fig. 5. Normalized antenna pattern processed by different methods.

eigendecomposition of $\mathbf{R}_{se}$. Therefore, its computational complexity can be expressed as $O(L \cdot N^3/8)$. Based on the above analysis, the total computational complexity of the proposed algorithm is $O(L \cdot N^2/4 + L^2 \cdot N^2/4 + 3N^2/4 + N^3/8 + L \cdot N^3/4)$.

IV. Experimental Results

This section conducts a comprehensive comparative analysis of the proposed method with the traditional algorithms, including BMP adaptive digital beamforming (BMP+ADBF), IBMP+ADBF, and 2-D frequency domain filtering [51], to verify the effectiveness and robustness of the proposed method. Finally, the simulation results of the point-like targets and the distributed scene are consistent, indicating that the proposed method can effectively suppress both 2D-ISRJ and NMM-ISRJ.

A. Algorithm Performance Analysis

1) BeamForming Pattern Analysis: In the beamforming pattern simulation, the signal-to-noise ratio (SNR) of the desired signal is set to 10 dB, the jamming-to-noise ratio (JNR) of the jamming signal is set to 15 dB, the angle of jamming is set to 30°, and the DOA estimation error of the desired signal is set to 2°. Fig. 5 shows the comparison of the beamforming pattern simulation results of the above four methods. Under the same experimental conditions, IBMP+ADBF lost half of its array DOFs, resulting in a significantly larger mainlobe width than BMP+ADBF and the proposed algorithm, reflecting the theoretical limitations of its beamforming performance. Further observation indicates that IBMP+ADBF has issues such as significantly elevated sidelobe level, shifted mainlobe position, and insufficient jamming zero notch. It is particularly worth noting that when there is a DOA estimation error of the desired signal, the beamforming pattern of IBMP+ADBF shows significant distortion, indicating that this method is extremely sensitive to angle estimation error. The performance of BMP+ADBF is also affected due to the one subarray DOFs loss, and it also shows problems such as the mainlobe shifted and the sidelobe level increased. BMP+ADBF is equally sensitive to the angle estimation error of the desired signal, resulting in a slight distortion of the beamforming pattern. Furthermore, the 2-D frequency domain filtering method is significantly affected by DOA estimation error and is most sensitive to the angle parameter. This leads to serious problems such as mainlobe broadening, mainlobe offset, sidelobe level rise, and insufficient jamming zero notch in Fig. 5. This further highlighted the performance limitations and poor robustness in the presence of DOA estimation error. In contrast, the proposed algorithm avoids the signal DOFs loss. As shown in Fig. 5, the proposed algorithm demonstrates the optimal comprehensive performance: the mainlobe width is the narrowest, the sidelobe level is the lowest, the pattern has no obvious distortion, and the mainlobe has no shift. Even in the case of DOA estimation error, the proposed algorithm can still maintain a relatively high mainlobe gain while forming a significantly deeper zero notch in the jamming direction.

Theoretical analysis indicates that the joint processing framework based on the proposed MBMP-CMSE and NC-MVDR can effectively achieve beamforming and zero notch. Under the condition of the coherent jamming and the estimation error wave arrival angle, the proposed algorithm still demonstrates the strong robustness, achieving effective jamming suppression while enhancing the desired signal.

2) Output SINR Performance Analysis: Fig. 6 systematically compares and evaluates the output SINR performance of BMP+ADBF, IBMP+ADBF, 2-D frequency domain filtering, and the proposed algorithm by 125 times Monte-Carlo simulation averaging experiments. Fig. 6(a) shows the curve of the output SINR varying with the desired signal DOA estimation error, where the DOA error range is set from 0° to 5°. Fig. 6(b) further analyzes the trend of the output SINR varying with the input SNR. Furthermore, to evaluate the performance of the proposed algorithm under different noise processing conditions, Fig. 7 compares the curves of the deviation between the output SINR and the optimal SINR of the proposed algorithm with the input SNR when noise is removed and when it is not removed.

According to the simulation result shown in Fig. 6(a), it can be seen that both BMP+ADBF, IBMP+ADBF, and 2-D frequency domain filtering exhibit sensitivity to DOA estimation error. In the presence of DOA mismatch, the output SINR of three algorithms decreased significantly, among which the performance deterioration of 2-D frequency domain filtering was particularly notable. In contrast, the IBMP+ADBF has a certain degree of robustness to DOA error. The decrease in its output SINR with the error increase is relatively small, and its overall performance is better than that of the BMP+ADBF and 2-D frequency domain filtering. Despite this, the output SINR of IBMP+ADBF under large DOA error remains at a relatively low level, indicating that it still has robustness limitations in practical application. Overall, the DOA estimation error has a significant impact on the performance of the traditional algorithm. In practical systems, the robustness design of the algorithm needs to be further considered. It can be clearly

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 02:12:53 UTC from IEEE Xplore. Restrictions apply.

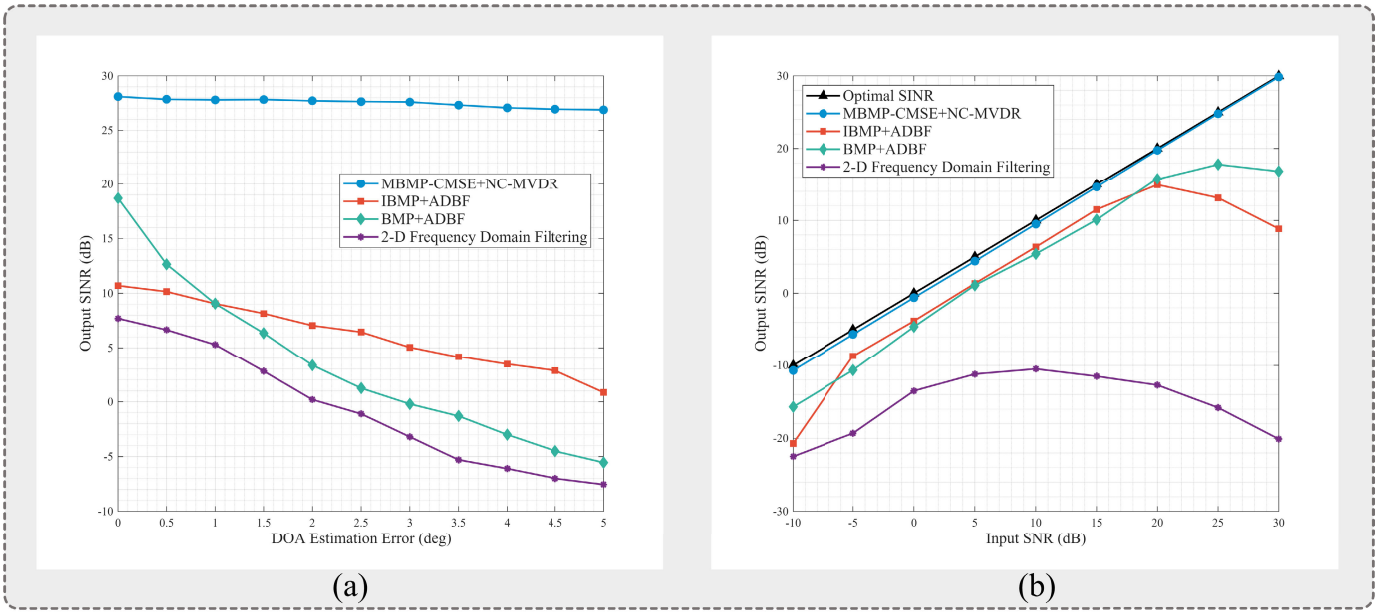


Fig. 6. Output SINR relationship curves of different blocking matrix preprocessing methods. (a) Relationship curves with DOA estimation error. (b) Relationship curves with the desired signal SNR.

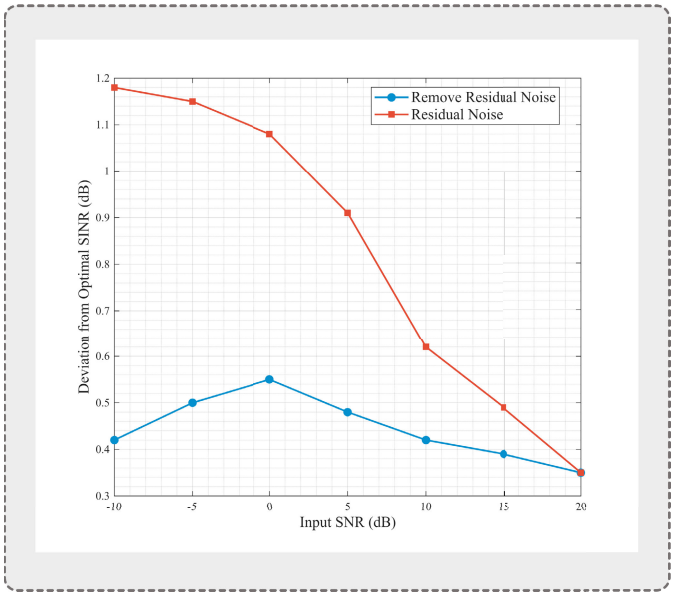


Fig. 7. Curve of the deviation optimal SINR with the desired signal SNR change.

observed from the simulation result that the proposed algorithm is significantly better than the traditional BMP+ADBF, IBMP+ADBF, and 2-D frequency domain filtering in output SINR. This result fully demonstrates that the adopted multistage blocking strategy can effectively enhance the robustness of the proposed algorithm under the DOA mismatch condition. Especially when the DOA error is relatively large, the proposed algorithm not only maintains a high output SINR level but also exhibits more stable performance. Meanwhile, compared with the IBMP+ADBF method, the output SINR improvement of the proposed algorithm also verifies the effectiveness of the covariance matrix extension transformation, indicating that the transformation has substantial contributions in applications.

Overall, the BMP and 2-D frequency domain filtering methods are extremely sensitive to angle estimation error, which can lead to a sharp decrease in SINR when DOA is mismatched. Although the IBMP method has good robustness against angle mismatched, its overall output SINR is low. In contrast, the output performance of the proposed algorithm is better, and multistage blocking can significantly improve the robustness of DOA estimation error, especially when the DOA error is large.

According to the simulation result shown in Fig. 6(b), it can be seen that within a wide range of input SNR, the output SINR of the proposed algorithm is always better than that of comparison algorithms, demonstrating a more outstanding signal enhancement capability. It is particularly noteworthy that the output SINR of the proposed algorithm shows a stable upward trend as the input SNR increases. Its value is closest to the theoretical optimal SINR curve, indicating that the proposed algorithm still has good robustness and consistency under the wide SNR condition.

According to the simulation result shown in Fig. 7, it can be seen that under the low input SNR condition, considering the influence of residual noise, the deviation between the output SINR of the proposed algorithm and the optimal SINR is smaller. As the input SNR gradually increases, the residual noise power is significantly lower than the desired signal power. At this time, the influence of residual noise on the covariance matrix estimation accuracy tends to weaken. Therefore, within the higher SNR range, the impact of whether residual noise is eliminated on the algorithm performance gradually decreases, and the output SINR under the conditions of removing and not removing noise tends to be consistent.

B. Point-Like Targets Simulation

This section first comparatively analyzes the jamming effects of the traditional mainlobe 2D-ISRJ and NMM-ISRJ

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 02:12:53 UTC from IEEE Xplore. Restrictions apply.

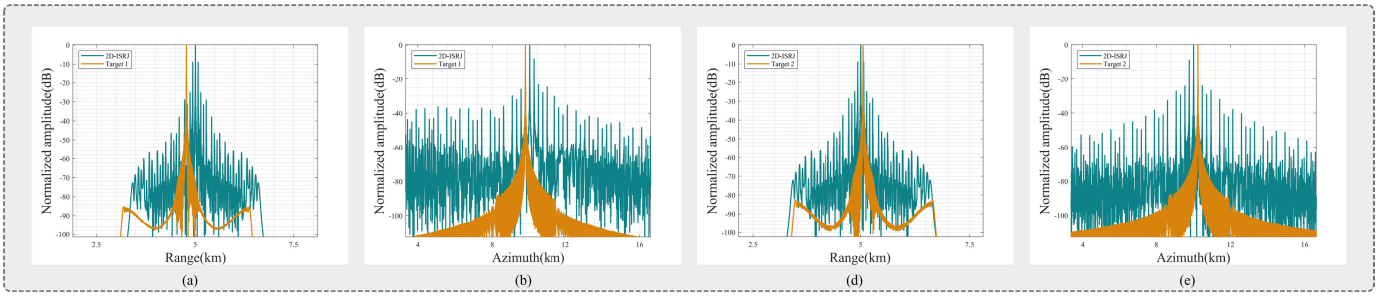


Fig. 8. 2D-ISRJ 1-D point-like targets simulations. (a) Target 1 and 2D-ISRJ 1-D range imaging result. (b) Target 1 and 2D-ISRJ 1-D azimuth imaging result. (c) Target 2 and 2D-ISRJ 1-D range imaging result. (d) Target 2 and 2D-ISRJ 1-D azimuth imaging result.

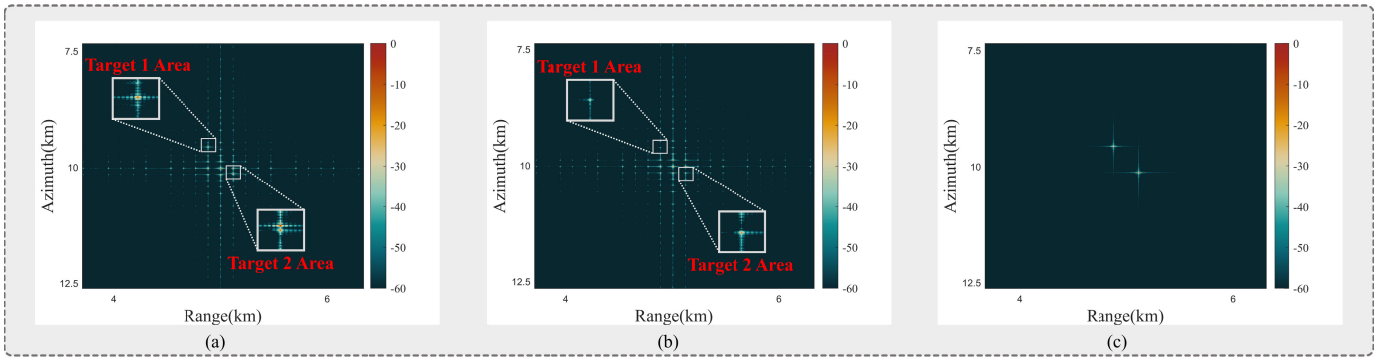


Fig. 9. Point-like targets imaging results of the proposed method suppress the mainlobe 2D-ISRJ. (a) Desired targets and 2D-ISRJ point-like targets imaging result. (b) Imaging result after MBMP-CMSE processing. (c) Imaging result after NC-MVDR processing.

TABLE I
POINT-LIKE TARGETS SIMULATION PARAMETERS

Parameters (Symbols)	Value
Reference Carrier Frequency (f_0)	5.4 GHz
Frequency Increment (Δf)	1562500 Hz
Number of T_x Elements (M)	8
Number of R_x Elements (N)	8
Element Spacing (d_a)	0.2 m
Pulse Duration (T_r)	60 μs
Platform Velocity (v_r)	200 m/s
Pulse Repetition Frequency (F_a)	550 Hz
Signal Bandwidth (B_r)	60 MHz
Target 1 position	(4874,9822) m
Target 2 position	(5061,1025) m
Signal-to-Noise Ratio	10 dB
Jamming-to-Noise Ratio	15 dB
DOA Estimation Error ($\Delta\alpha$)	2°

TABLE II
2D-ISRJ SIMULATION PARAMETERS

Parameters (Symbols)	Value
Range Interrupted Sampling Pulse Width(T_{w1})	0.72 μs
Range Interrupted Sampling Repetition Period (T_{s1})	1.2 μs
Range Duty Ratio	0.6
Range Sampling Times	50
Azimuth Interrupted Sampling Pulse Width(T_{w2})	0.0120 s
Azimuth Interrupted Sampling Repetition Period (T_{s2})	0.0199 s
Azimuth Duty Ratio	0.6
Azimuth Sampling Times	20

TABLE III
NMM-ISRJ SIMULATION PARAMETERS

Parameters (Symbols)	Value
Number of Random False Targets	200
Fast Time Bandwidth (B_{nr})	0.6 MHz
Slow Time Bandwidth (B_{na})	282.01 Hz

on the FDA-MIMO SAR system, and presents the point-like targets simulation results after jamming suppression by the proposed algorithm. The simulation parameters of the point-like targets, 2D-ISRJ, and NMM-ISRJ correspond to the specific parameters listed in Tables I–III, respectively. In the simulation scenario configuration, the jammer is positioned at the geometric center of the observation scene. In the simulation experiments of point-like targets, by comparing the jamming characteristics and suppression effects of the traditional 2D-ISRJ and NMM-ISRJ, the effectiveness of the proposed combined MBMP-CMSE and NC-MVDR method can be verified.

First, for the traditional 2D-ISRJ, Fig. 8(a)–(d) presents the 1-D range and azimuth imaging results of target 1 and target 2. It can be observed from these figures that the false targets generated by 2D-ISRJ exhibit a uniform spacing distribution in both the range and azimuth dimensions, with strong consistency in amplitude, forming a regular grid-like array of false targets. This regularity makes it easier to identify the real targets, but it also exposes the disadvantages of 2D-ISRJ, such as easy identification and reduced deception. Fig. 9(a) shows the 2-D point-like targets imaging results of the real targets and 2D-ISRJ. As shown in Fig. 9(b), after being

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 02:12:53 UTC from IEEE Xplore. Restrictions apply.

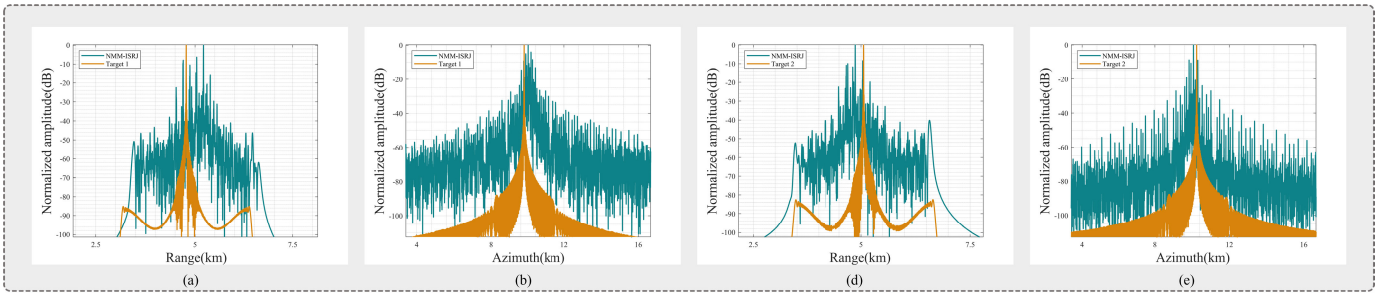


Fig. 10. NMM-ISRJ 1-D point-like targets simulations. (a) Target 1 and NMM-ISRJ 1-D range imaging result. (b) Target 1 and NMM-ISRJ 1-D azimuth imaging result. (c) Target 2 and NMM-ISRJ 1-D range imaging result. (d) Target 2 and NMM-ISRJ 1-D azimuth imaging result.

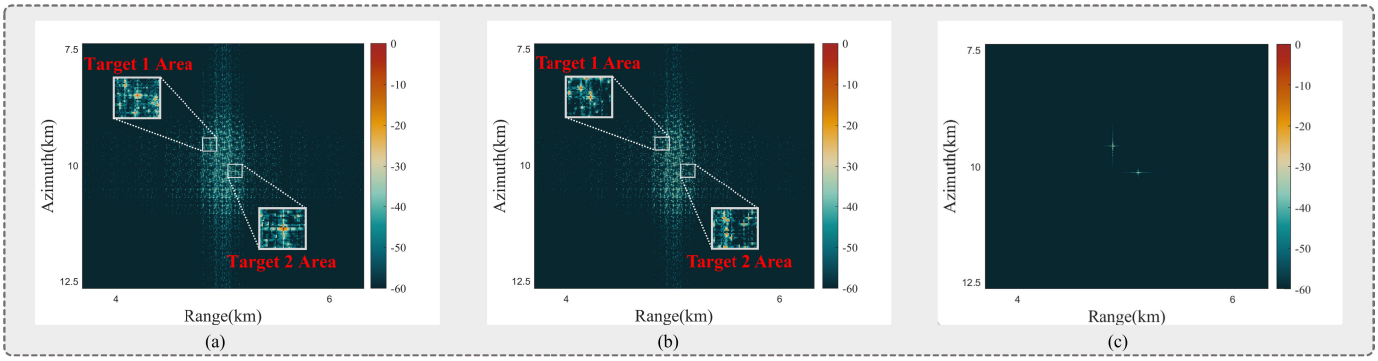


Fig. 11. Point-like targets imaging results of the proposed method suppress the mainlobe NMM-ISRJ. (a) Desired targets and NMM-ISRJ point-like targets imaging result. (b) Imaging result after MBMP-CMSE processing. (c) Imaging result after NC-MVDR processing.

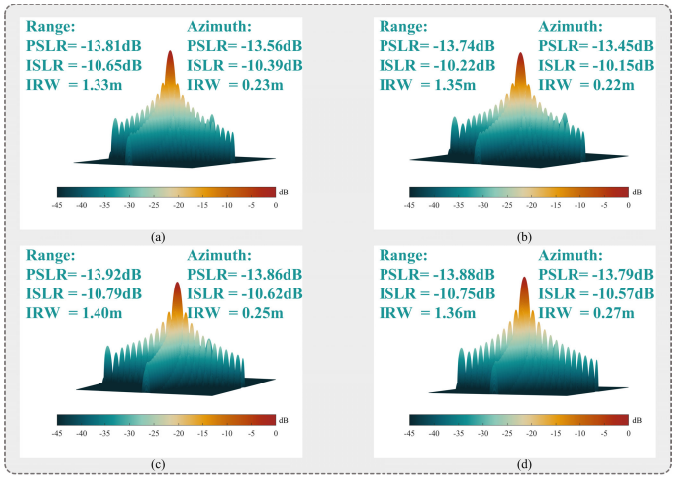


Fig. 12. Point-like targets 3-D response simulations. (a) Target 1 3-D response in Fig. 9(c). (b) Target 2 3-D response in Fig. 9(c). (c) Target 1 3-D response in Fig. 11(c). (d) Target 2 3-D response in Fig. 11(c).

processed by MBMP-CMSE, even in the presence of DOA estimation errors, the desired signal can be effectively blocked, while only the jamming and noise components are retained. In Fig. 9(c), further through NC-MVDR beamforming, the 2D-ISRJ signal is significantly suppressed, proving that the proposed method can effectively suppress 2D-ISRJ.

In contrast, Fig. 10(a)–(d) shows the 1-D range and azimuth imaging results of target 1 and target 2 under the influence of NMM-ISRJ. NMM-ISRJ generated dense false targets with uneven spacing distribution and random amplitudes through the offline noise modulation template. This jamming pattern



Fig. 13. Distributed scene simulation template.

effectively avoids the regularity of traditional deceptive or occlusive false targets, and its highly random characteristic significantly enhances the realism and concealment of NMM-ISRJ. In Fig. 11(a), the 2-D point-like targets imaging results visually demonstrate that the true targets can be completely submerged in the false targets generated by NMM-ISRJ, making it difficult to distinguish the true targets. However, even when the true targets are difficult to distinguish, and there is DOA estimation error, the proposed MBMP-CMSE method [in Fig. 11(b)] can still exhibit strong robustness, being able to accurately block the desired signal and retain the jamming and noise information. Finally, in Fig. 11(c), after being processed by NC-MVDR, NMM-ISRJ can be effectively suppressed, fully demonstrating the effectiveness of the proposed method in dealing with irregular mainlobe composite jamming.

To quantitatively evaluate the focusing performance of point-like targets after jamming suppression, Fig. 12 shows the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 02:12:53 UTC from IEEE Xplore. Restrictions apply.

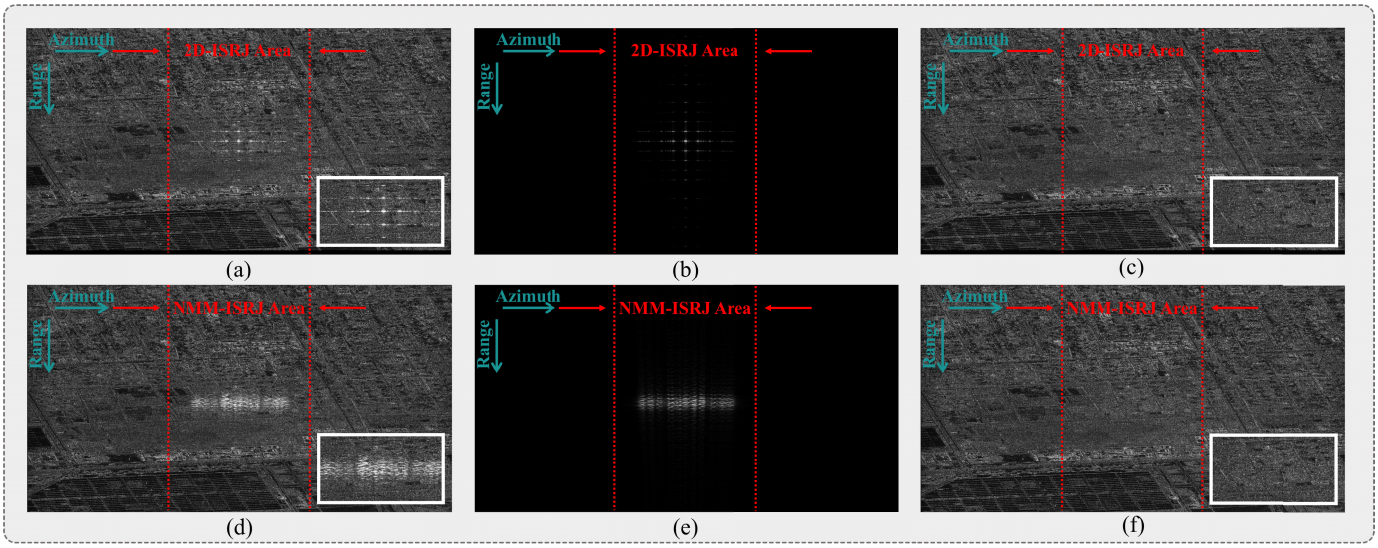


Fig. 14. Distributed scene simulation results of the proposed anti-jamming method. (a) Distributed scene and 2D-ISRJ output result. (b) Distributed scene and 2D-ISRJ output result after MBMP-CMSE. (c) Distributed scene and 2D-ISRJ output result after NC-MVDR. (d) Distributed scene and NMM-ISRJ output result. (e) Distributed scene and NMM-ISRJ output result after MBMP-CMSE. (f) Distributed scene and NMM-ISRJ output result after NC-MVDR.

key performance indicators, including the peak sidelobe ratio (PSLR), the integrated sidelobe ratio (ISLR), and the impulse response width (IRW) [52]. Fig. 12(a) and (b) shows the point-like target responses of target 1 and target 2 after suppressing 2D-ISRJ, while Fig. 12(c) and (d) presents the results after suppressing NMM-ISRJ. From the quantitative indicators provided in Fig. 12, it can be seen that after processing with the proposed method, the point target responses of both targets have achieved good focusing characteristics, and the values of PSLR, ISLR, and IRW are all within the theoretical range. This indicates that the proposed method effectively suppresses the jamming energy while ensuring excellent imaging quality.

In summary, the proposed processing framework, which combines MBMP-CMSE with NC-MVDR, not only effectively counteracts the regular traditional 2D-ISRJ, but also demonstrates outstanding suppression capabilities for more deceptive and cover-based NMM-ISRJ. The proposed method significantly enhances the imaging ability of FDA-MIMO SAR in the complex electromagnetic mainlobe jamming environment while ensuring good focusing performance of the desired targets.

C. Distributed Scene Simulation

To comprehensively and objectively evaluate the overall performance of the proposed method in a realistic environment, this section conducts simulation experiments in a distributed scene. The system parameters involved in the distributed scene simulation are detailed in Table IV, and the size of the distributed scene template is 1300×700 m. In Fig. 13, it is assumed that the center of the distributed scene is the origin of the 2-D plane coordinate system, with the upward direction as the positive y-axis and the rightward direction as the positive x-axis. The jammer is set at (220, 150, 0) m. The 2D-ISRJ and NMM-ISRJ simulation parameters used in the distributed scene simulations still follow those from Tables II and III, respectively. Fig. 13 shows the distributed scene template used in this simulation, which covers various types of ground features with typical characteristics. The structure is rich and the details are clear, enabling a thorough simulation of the complex environment in actual SAR observations. The distributed scene simulation experiments can more comprehensively verify the effectiveness of the proposed method.

TABLE IV
DISTRIBUTED SCENE SIMULATION PARAMETERS

Parameters (Symbols)	Value
Reference Carrier Frequency (f_0)	9.6 GHz
Frequency Increment (Δf)	400 kHz
Number of T_x Elements (M)	8
Number of R_x Elements (N)	8
Element Spacing (d_a)	0.2 m
Pulse Duration (T_r)	10 μs
Platform Velocity (v_r)	200 m/s
Platform Height (H)	5.7 km
Slant Range (R_t)	8.5 km
Pulse Repetition Frequency (F_a)	1000 Hz
Signal Bandwidth (B_r)	120 MHz
DOA Estimation Error ($\Delta\alpha$)	2°
Jammer Position ($x_j, y_j, z_j,$)	(220, 150, 0) m

From the perspective of jamming characteristics, Fig. 14(a) and (d) clearly demonstrate the essential differences between the traditional 2D-ISRJ and NMM-ISRJ. The false targets generated by the traditional 2D-ISRJ exhibit regular patterns of uniform spacing and consistent amplitude in spatial distribution, making them relatively easy to identify and suppress. In contrast, NMM-ISRJ can flexibly generate false targets with both deceptive and dense covering properties through offline design of the noise modulation template. Its jamming form is highly variable and more closely resembles practical engineering applications. Therefore, the NMM-ISRJ has stronger

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 02:12:53 UTC from IEEE Xplore. Restrictions apply.

jamming concealment capabilities, significantly increasing the suppression difficulty and posing greater challenges to the radar system.

In terms of the algorithm processing effect, Fig. 14(b) and (e) indicates that after being processed by MBMP-CMSE, the desired signal is effectively suppressed, and the output results only contain the jamming and noise information, laying a good foundation for subsequent processing. More importantly, as shown in Fig. 14(c) and (f), after further processing by the NC-MVDR beamformer, the proposed method not only can effectively suppress the traditional 2D-ISRJ with regularity, but also can achieve a good jamming suppression effect when facing NMM-ISRJ with a more composite jamming form and more difficult to suppress.

To evaluate the suppression effect of false targets, this article uses the electronic counter measures improvement factor (EIF) as the evaluation factor, i.e.,

$$\mathrm{EIF} = \frac{(J/S)}{(J/S)'} \tag{62}$$

where (J/S) represents the jamming-to-signal ratio (JSR) of the radar system output when no anti-jamming measures are taken, and $(J/S)'$ is the JSR of the radar system output after adopting anti-jamming measures. The larger the EIF value, the better the anti-jamming performance of the radar system after taking anti-jamming measures. After being processed by the method proposed, the EIF value of the anti-2D-ISRJ is about 28.52 dB, and the EIF value of the anti-NMM-SRJ is about 28.51 dB. This indicates that the proposed method can effectively suppress false targets without affecting the desired signal imaging.

The distributed scene experimental results show that the proposed joint processing method can not only effectively suppress the traditional 2D-ISRJ with obvious patterns, but also exhibits good suppression effects on NMM-ISRJ with more difficult to handle. Under both jamming modes, the proposed method successfully restored the distributed targets scene and significantly enhanced the anti-jamming ability of FDA-MIMO SAR in a complex electromagnetic environment.

V. Conclusion

This article addresses the significant challenge of NMM-ISRJ composite jamming suppression in the FDA-MIMO SAR system and proposes an innovative joint processing solution. The proposed method nearly completely suppresses the desired signal through MBMP when there is a DOA estimation error. On this basis, the CMSE technology is proposed to effectively compensate for the array DOFs loss caused by the multistage blocking operation. Finally, combined with the NC-MVDR beamforming algorithm, the overall robustness of the proposed method is significantly enhanced. Quantitative analysis experiments show that the proposed method significantly outperforms the traditional BMP, IBMP, and 2-D frequency domain filtering methods in terms of beam pattern performance and output SINR. Point-like targets and distributed scene simulation experiments show that the proposed method can effectively suppress the mainlobe 2D-ISRJ and NMM-ISRJ, providing an efficient and reliable technical approach for radar mainlobe anti-jamming in a complex electromagnetic environment.

Appendix A

Considering the noise template $n(t_r, t_a)$, it can be modeled as the composition of multiple single-frequency components in the frequency domain, i.e.,

$$n(t_r, t_a) = n(t_r) \otimes n(t_a) \tag{A1}$$

where $n(t_r)$ and $n(t_a)$ denote the fast time and the slow time noise template, respectively. The modulation principle of $n(t_r)$ and $n(t_a)$ are similar to the frequency shifting jamming technology, that is, $n(t_r)$ and $n(t_a)$ achieve spectrum shifting by controlling the switching of carrier frequencies between discrete frequency points. Specifically, $n(t_r)$ and $n(t_a)$ can be expressed as, respectively,

$$n(t_r) = \sum_{f_{i,r}=-B_{nr}}^{B_{nr}} \exp\left(j2\pi f_{i,r} t_r\right) \tag{A2}$$

$$(t_a) = \sum_{f_{i,a}=-B_{na}}^{B_{na}} \exp\left(j2\pi f_{i,a} t_a\right) \tag{A3}$$

where B_{nr} and B_{na} represent the fast time and the slow time bandwidth, respectively. The imaging result of the noise template can be approximated as a series of impulse functions synthesized with the jammer as the center. The positions of these impulses are determined by the frequency points of the noise template, while the jamming coverage range (area) is determined by B_{nr} and B_{na} together. From this, it can be seen that the frequency domain design of $n(t_r, t_a)$ is the core factor in controlling and optimizing jamming effects [11], [53].

The specific generation process of $n(t_r, t_a)$ can be briefly summarized as follows: first, performing fast Fourier transform (FFT) on the initial noise template $n_0(t_r, t_a)$ to obtain its 2-D frequency domain representation $n_0(f_r, f_a)$. Second, based on the preset jamming range, low-pass filters are constructed for both the fast and slow time dimensions. The frequency domain noise template $n_0(f_r, f_a)$ can be filtered accordingly, i.e.,

$$n_1(f_r, f_a) = n_0(f_r, f_a) H(f_r) H(f_a) \tag{A4}$$

where $H(f_r)$ and $H(f_a)$ represent the fast time and the slow time low-pass filter, respectively. By processing the modulation template through the above step, a suppression surface centered around the jammer can be generated.

Then, the frequency domain sampling function can be expressed as

$$S(f_r, f_a) = \begin{cases} 1, & (f_r, f_a) \in \mathcal{Q} \\ 0, & (f_r, f_a) \notin \mathcal{Q} \end{cases} \tag{A5}$$

where $\mathcal{Q}$ denotes a series of specific frequency points in the frequency domain, which is a predefined set. After the above step of processing, the false targets can be generated at the designated location. Performing frequency domain sampling on $n_1(f_r, f_a)$, i.e.,

$$n(f_r, f_a) = n_1(f_r, f_a) S(f_r, f_a). \tag{A6}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 02:12:53 UTC from IEEE Xplore. Restrictions apply.

Finally, the filtered frequency domain noise signal $n(f_r, f_a)$ is transformed back to the time domain by inverse FFT, and the output result is the time domain noise template $n(t_r, t_a)$. This modulation template can be pregenerated offline by the jammer, thus the jammer has ample computation time. In practical applications, when the jammer modulates the intercepted radar signal, only one time domain multiplication operation is required, and the process is extremely simple and efficient.

APPENDIX B

From Appendix A, it can be concluded that in the modulation template, the relative position of the false targets can be achieved by setting the specified frequency points position. Therefore, the change in the frequency points' relative position can affect the range and azimuth offsets of the false targets, which can be expressed as, respectively,

$$f_{i,r} = -\frac{2K_r R_{i,f}}{c} \tag{B1}$$

$$f_{i,a} = \frac{2v_r A_{i,f}}{\lambda R_0} \tag{B2}$$

where $f_{i,r}$ and $f_{i,a}$ represent the fast time and slow time frequency position of the false targets, respectively. $R_{i,f}$ and $A_{i,f}$ are the range and azimuth position of the false targets relative to the jammer, respectively. Before implementing multiplication modulation on intercepted radar signal, the jammer requires key reconnaissance parameters, among which the key parameters that need to be analyzed in detail include relative errors such as chirp rate error ε_{K_r}, velocity error ε_{v_r}, wavelength error ε_λ, and initial slant range error ε_{R_0} [11], [53].

According to (B1) and (B2), it can be seen that in the case where the jammer has reconnaissance errors, the specified frequency points position of the false targets in the modulation template is, respectively,

$$f_{i,r}^{'} = \left(1 + \varepsilon_{K_r}\right) f_{i,r} \tag{B3}$$

$$f_{i,a}^{'} = \frac{\left(1 + \varepsilon_{v_r}\right)}{\left(1 + \varepsilon_\lambda\right)\left(1 + \varepsilon_{R_0}\right)} f_{i,a}. \tag{B4}$$

Therefore, the range and azimuth position of the false targets can be expressed as, respectively,

$$y_f = y_j + \frac{c\tau_{jm}}{2} - \frac{cm\, f_{s1}}{2K_r} - \frac{c f_{i,r}}{2K_r}\left(1 + \varepsilon_{K_r}\right) \tag{B5}$$

$$x_f = x_j - \frac{v_r n f_{s2}}{K_a} + \frac{v_r f_{i,a}}{K_a} \frac{\left(1 + \varepsilon_{v_r}\right)}{\left(1 + \varepsilon_\lambda\right)\left(1 + \varepsilon_{R_0}\right)} \tag{B6}$$

where K_a is the azimuth chirp rate. Based on the above theoretical derivation, it can be seen that when there is a reconnaissance parameter estimation error in the jammer, there will be a deviation between the jamming imaging result and the preset position of the false targets, but it will not affect the output effect of the jamming. Among them, the range offset position of the false target is affected by ε_{K_r}, and the larger ε_{K_r}, the greater the range offset position. The azimuth offset position is affected by ε_{v_r}, ε_λ, and ε_{R_0}, which are proportional to ε_{v_r} and inversely proportional to ε_λ and ε_{R_0} [11].

REFERENCES

[1] D. Zhu et al., "An extended two step approach to high-resolution airborne and spaceborne SAR full-aperture processing," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 10, pp. 8382–8397, Oct. 2021.

[2] G. Jin et al., "New insights into SAR alternate transmitting mode based on waveform diversity," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5209209.

[3] G. Jin, W. Wang, Y. Deng, H. Yan, and R. Wang, "A novel range-azimuth joint modulation scheme for range ambiguity suppression," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5207210.

[4] G. Jin, X. Zhang, J. Huang, and D. Zhu, "High freedom parameterized FM (HFPFM) code: Model, correlation function, and advantages," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 60, no. 5, pp. 6284–6298, Oct. 2024.

[5] T. Zhou, G. Jin, D. Liang, P. Lu, and D. Zhu, "A novel phase synchronization method for spaceborne multistatic SAR," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5200814.

[6] N. Li and X. Hu, "Ultrawideband mutual RFI mitigation between SAR satellites: From the perspective of European Sentinel-1A," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5229020.

[7] D. Cheng et al., "2-D precise controllable HRWS-SAR smart jamming: Through segmented-phase modulation," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5213720.

[8] Y. Fan, A. Wen, Y. Men, and Y. Wang, "A photonic-assisted 2-D ISRJ jamming strategy for SAR with self-interference cancellation," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 61, no. 5, pp. 11919–11929, Oct. 2025.

[9] X. Wang, J. Liu, W. Zhang, Q. Fu, Z. Liu, and X. Xie, "Mathematic principles of interrupted-sampling repeater jamming (ISRJ)," *Sci. China Ser. F, Inf. Sci.*, vol. 50, no. 1, pp. 113–123, Feb. 2007.

[10] X. Wu, X. Wang, and H. Lu, "Study of intermittent sampling repeater jamming to SAR," *J. Astronaut*, vol. 30, pp. 2043–2048, Feb. 2009.

[11] G. Sun, S. Xing, D. Huang, Y. Li, and X. Wang, "Jamming method of intermittent sampling against SAR-GMTI based on noise multiplication modulation," *J. Syst. Eng. Electron*, vol. 44, no. 10, pp. 3071–3509, Oct. 2022.

[12] K. Zhou, D. Li, S. Quan, T. Liu, Y. Su, and F. He, "SAR waveform and mismatched filter design for countering interrupted-sampling repeater jamming," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5214514.

[13] K. Zhou, Y. Su, D. Wang, H. Ma, L. Liu, and C. Li, "Improved SAR interrupted-sampling repeater jamming countermeasure based on waveform agility and mismatched filter design," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5206916.

[14] Z. Shi, H. Wu, X. Zhu, W. Wang, J. Zhang, and Q. Li, "A notch LFM waveform design method for anti-interrupted sampling repeater jamming," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 61, no. 5, pp. 11325–11337, Oct. 2025.

[15] H. Su, Q. Bao, J. Pan, F. Guo, and W. Hu, "Waveform-domain complementary signal sets for interrupted sampling repeater jamming suppression," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 60, no. 5, pp. 7142–7158, Oct. 2024.

[16] Y. Zhang, Y. Wei, and L. Yu, "ISRJ rejection with MIMO radar using space–time complex modulation in multijamming scenario," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 59, no. 6, pp. 8725–8742, Dec. 2023.

[17] S. Gong, X. Wei, and X. Li, "ECCM scheme against interrupted sampling repeater jammer based on time-frequency analysis," *J. Syst. Eng. Electron.*, vol. 25, no. 6, pp. 996–1003, Dec. 2014.

[18] J. Wei et al., "A time-domain filtering method based on intrapulse joint interpulse coding to counter interrupted sampling repeater jamming in SAR," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5215912.

[19] Y. Meng, L. Yu, and Y. Wei, "Interrupted sampling repeater jamming suppression based on time-frequency segmentation network and target signal reconstruction," in *Proc. IEEE Int. Radar Conf. (RADAR)*, Sydney, NSW, Australia, Nov. 2023, pp. 1–5.

[20] L. Lu and M. Gao, "An improved sliding matched filter method for interrupted sampling repeater jamming suppression based on jamming reconstruction," *IEEE Sensors J.*, vol. 22, no. 10, pp. 9675–9684, May 2022.

[21] Y. Gao, L. Ren, Q. Deng, H. Fan, Q. Liu, and E. Mao, "Refined time-frequency feature matching for interrupted sampling repeater jamming suppression," in *Proc. IEEE Int. Conf. Signal, Inf. Data Process. (ICSIDP)*, Zhuhai, China, Nov. 2024, pp. 1–5.

[22] F. Chen, R. Li, L. Ding, L. Liu, L. Dai, and G. Deng, "A method against DRFM dense false target jamming based on jamming recognization," in *Proc. IET Int. Radar Conf.*, Oct. 2015, pp. 1–4.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 02:12:53 UTC from IEEE Xplore. Restrictions apply.

[23] W.-Q. Wang, M. Dai, and Z. Zheng, "FDA radar ambiguity function characteristics analysis and optimization," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 54, no. 3, pp. 1368–1380, Jun. 2018.
[24] F. Wan, J. Xu, Y. Xu, W. Wang, G. Liao, and Y. Zhang, "Clutter suppression for spaceborne FDA-MIMO radar with QPC," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 62, pp. 661–681, 2026.
[25] S. Shao, W.-Q. Wang, Y. Sun, and S. Zhang, "Effects of spatial position on mainlobe interference suppression performance of FDA-MIMO radar," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 60, no. 6, pp. 9437–9443, Dec. 2024.
[26] M. Tan, C. Wang, and Z. Li, "Correction analysis of frequency diverse array radar about time," *IEEE Trans. Antennas Propag.*, vol. 69, no. 2, pp. 834–847, Feb. 2021.
[27] M. Tan and C. Wang, "Reply to comments on, 'Correction analysis of 'Frequency diverse array radar about time,'" *IEEE Trans. Antennas Propag.*, vol. 71, no. 3, pp. 2899–2902, Mar. 2023.
[28] J. Xu, Y. Zhang, G. Liao, Y. Xu, and W. Wang, "Angle-dependent matched filtering for moving target indication with coherent FDA radar," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 60, no. 3, pp. 3523–3536, Jun. 2024.
[29] J. Qi, L. Lan, and G. Liao, "FDA-MIMO transceiver design for deceptive jammer suppression," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 61, no. 2, pp. 4394–4408, Apr. 2025.
[30] L. Lan, M. Rosamilia, A. Aubry, A. D. Maio, and G. Liao, "Single-snapshot angle and incremental range estimation for FDA-MIMO radar," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 57, no. 6, pp. 3705–3718, Dec. 2021.
[31] Z. Qiu, K. Duan, X. Yang, and Y. Wang, "Range-ambiguous clutter suppression for space-based early warning radar via EPC–FDA–MIMO with nonorthogonal waveforms," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 60, no. 5, pp. 7106–7124, Oct. 2024.
[32] J. Xu, G. Liao, S. Zhu, L. Huang, and H. C. So, "Joint range and angle estimation using MIMO radar with frequency diverse array," *IEEE Trans. Signal Process.*, vol. 63, no. 13, pp. 3396–3410, Jul. 2015.
[33] X. Wang et al., "A novel FDA-MIMO SAR mainlobe jamming suppression method utilizing modified STSO-waveform digital beamforming," *IEEE Trans. Geosci. Remote Sens.*, vol. 64, 2026, Art. no. 5204116.
[34] H. Zhang et al., "A fast repeater mainlobe deceptive jamming suppression method for FDA-MIMO SAR under complex motion condition," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 62, pp. 862–878, 2026.
[35] P. Antonik, M. C. Wicks, H. D. Griffiths, and C. J. Baker, "Frequency diverse array radars," in *Proc. IEEE Radar Conf.*, Verona, NY, USA, Apr. 2006, pp. 215–217.
[36] G. Krieger, "MIMO-SAR: Opportunities and pitfalls," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 5, pp. 2628–2645, May 2014.
[37] G. Jin, Y. Wang, D. Zhu, S. Niu, and H. Yan, "A reconfigurable MIMO-SAR transmission scheme based on inter-pulse and intra-pulse joint phase modulation," *IEEE Trans. Signal Process.*, vol. 70, pp. 4265–4276, 2022.
[38] Y. Wang et al., "A novel MIMO-SAR echo separation solution for reducing the system complexity: Spectrum preprocessing and segment synthesis," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5206517.
[39] L. Lan, Y. Zhang, J. Xu, G. Liao, and H. C. So, "Suppressing mainlobe deceptive jammers via two-low-rank matrix decomposition in FDA-MIMO radar," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 61, no. 2, pp. 2885–2898, Apr. 2025.
[40] L. Lan, M. Rosamilia, A. Aubry, A. De Maio, and G. Liao, "FDA-MIMO transmitter and receiver optimization," *IEEE Trans. Signal Process.*, vol. 72, pp. 1576–1589, 2024.
[41] Y. Cheng, D. Zhu, G. Jin, J. Zhang, S. Niu, and Y. Wang, "A novel intrapulse repeater mainlobe-jamming suppression method with MIMO-SAR," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.
[42] M. Lou, J. Yang, Z. Li, H. Ren, H. An, and J. Wu, "Joint optimal and adaptive 2-D spatial filtering technique for FDA-MIMO SAR deception jamming separation and suppression," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5238414.
[43] L. Lan, G. Liao, J. Xu, Y. Zhang, and B. Liao, "Transceive beamforming with accurate nulling in FDA-MIMO radar for imaging," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 6, pp. 4145–4159, Jun. 2020.
[44] M. Tang, D. Qi, C. Liu, and Y. Zhao, "New adaptive beamformer for coherent interference based on multistage blocking," *J. Electron. Inf. Techn*, vol. 41, no. 7, pp. 1–16, Jul. 2019.
[45] S. Ma and F. Pan, "Symmetric extension of steering vectors and beamforming," *Prog. Electromagn. Res. M*, vol. 76, pp. 19–29, 2018.
[46] J. Li, P. Stoica, and Z. Wang, "Doubly constrained robust Capon beamformer," *IEEE Trans. Signal Process.*, vol. 52, no. 9, pp. 2407–2423, Sep. 2004.
[47] W. Chen, W. Xie, and Y. Wang, "Range-dependent ambiguous clutter suppression for airborne SSF-STAP radar," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 58, no. 2, pp. 855–867, Apr. 2022.
[48] Y. Wang, X. Pan, G. Jin, T. Zhou, D. Wu, and D. Zhu, "LASSO regression-based DBF technique for waveform decoupling of MIMO-SAR with nonlinear array configuration," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5219718.
[49] S. Niu, G. Jin, Y. Cheng, Y. Wang, H. Zhang, and D. Zhu, "A novel MIMO SAR transmission scheme for restoring repeated equivalent phase centers," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5216618.
[50] M. Wang, S. Guo, X. Cong, Y. Wang, and W. Sheng, "A joint anti-jamming algorithm in space-domain of mainlobe and sidelobe for mainlobe shape-preserving," *Modern Radar*, vol. 47, no. 1, pp. 71–78, 2025.
[51] J. Xu, G. Liao, Y. Zhang, and Q. Su, "On anti-jamming technique with waveform diverse array radar," *Acta Electron. Sin*, vol. 47, no. 3, pp. 545–551, 2019.
[52] I. G. Cumming and F. H. Wong, *Digital Processing of Synthetic Aperture Radar Data: Algorithms and Implementation*. Norwood, MA, USA: Artech House, 2005.
[53] D. Huang, S. Xing, Y. Li, Y. Liu, and S. Xiao, "Smart jamming method against SAR based on multiplication modulation," *J. Syst. Eng. Electron.*, vol. 43, no. 11, pp. 3160–3168, 2021.



Hanqing Zhang received the B.S. degree in computer science and technology and the M.S. degree in computer architecture from Henan University, Kaifeng, China, in 2019 and 2022, respectively. He is currently pursuing the Ph.D. degree in electronic information with the Key Laboratory of Radar Imaging and Microwave Photonics, Ministry of Education, Nanjing University of Aeronautics and Astronautics (NUAA), Nanjing, China.

Since 2024, he has been working as a Joint Trainee with the AVIC Leihua Electronic Technology Research Institute, Wuxi, China, under the joint supervision of Prof. D. Zhu, S. Ye, and H. Zhang. His research interests include synthetic aperture radar (SAR) signal processing and imaging, frequency diverse array multiple-input multiple-output (FDA-FMIMO) SAR signal processing, and anti-jamming.



Guodong Jin (Senior Member, IEEE) received the B.S. degree in electronic science and technology from Lanzhou University, Lanzhou, China, in 2015, and the Ph.D. degree in communication and information systems from the University of Chinese Academy of Sciences (UCAS), Beijing, China, in 2020, under the supervision of Prof. Y. Deng and R. Wang.

From 2015 to 2020, he was at the Department of Space Microwave Remote Sensing System, Institute of Electronics, Chinese Academy of Sciences (IECAS), Beijing. From 2019 to 2020, he was a Visiting Ph.D. Student at the Universitá degli Studi di Naples Federico II, Naples, Italy, under the local supervision of Prof. A. De Maio and Prof. A. Aubry. Since 2020, he has been an Associate Researcher with the College of Electronic and Information Engineering, Nanjing University of Aeronautics and Astronautics (NUAA), Nanjing, China. His main research interests include waveform design/optimization, multiple-input multiple-output (MIMO) synthetic aperture radar (SAR), SAR ambiguity suppression, as well as SAR data processing and applications.

Dr. Jin received the Special Prize of the President of the Chinese Academy of Sciences in 2020.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 02:12:53 UTC from IEEE Xplore. Restrictions apply.



Hongbo Zhang received the B.S. degree in electronic information engineering from Hunan University (HNU), Changsha, China, in 2005, and the M.S. degree in signal and information processing from Beihang University (BUAA), Beijing, China, in 2008.

Since 2008, she has been working with the AVIC Leihua Electronic Technology Research Institute, Wuxi, China. Her research interests include radar system design, high-resolution and wide-swath synthetic aperture radar (HRWS-SAR), SAR target location, SAR-ground moving target indication (SAR-GMTI), and ground moving target tracking.



Shilin Niu received the B.S. degree in computer science and technology and the M.S. degree in computer architecture from Henan University, Kaifeng, China, in 2016 and 2020, respectively, and the Ph.D. degree in communication and information systems with the Key Laboratory of Radar Imaging and Microwave Photonics, Ministry of Education, Nanjing University of Aeronautics and Astronautics (NUAA), Nanjing, China, in 2024.

Since 2025, he has been with the Network Operation Department of China United Network Communications Company Ltd., Henan Branch, Zhengzhou, China. His research interests include synthetic aperture radar (SAR) data processing and imaging, multiple-input multiple-output (MIMO) SAR signal processing, waveform design, range ambiguity suppression, low-altitude intelligent networking, integrated sensing and communication (ISAC), artificial intelligence, next-generation internet, and 6G wireless networks.



Jie Lv received the B.S. degree in electrical engineering and automation from Hebei University of Science and Technology, Shijiazhuang, China, in 2012, and the M.S. degree in navigation, guidance, and control from Harbin Engineering University, Harbin, China, in 2015.

He is currently working with the AVIC Leihua Electronic Technology Research Institute, Wuxi, China. His research interests include the design of airborne radar systems for sea search.



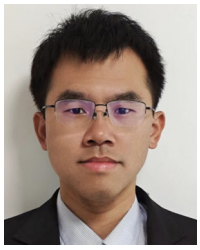
Yuan Cheng received the B.S. degree in electronic engineering from Nanjing University of Information Science and Technology (NUIST), Nanjing, China, in 2014, the M.S. degree in electronics from China Ship Research and Development Academy, Beijing, China, in 2018, and the Ph.D. degree in communication and information systems from the Key Laboratory of Radar Imaging and Microwave Photonics, Ministry of Education, Nanjing University of Aeronautics and Astronautics (NUAA), Nanjing, in 2022.

He is currently working with the AVIC Leihua Electronic Technology Research Institute, Wuxi, China. His research interests include radar signal processing and its applications.



Jing Ding received the B.S. degree in information engineering and the M.S. degree in signal and information processing from Nanjing University of Aeronautics and Astronautics (NUAA), Nanjing, China, in 2014 and 2017, respectively.

She is currently working with the AVIC Leihua Electronic Technology Research Institute, Wuxi, China. Her research interests include synthetic aperture radar (SAR) motion compensation, SAR imaging, and high forward-squint SAR.



Zhenyu Guo received the B.S. degree in electronics and information engineering and the Ph.D. degree in signal and information processing from the University of Chinese Academy of Sciences (UCAS), Beijing, China, in 2010 and 2017, respectively.

He is currently working with the AVIC Leihua Electronic Technology Research Institute, Wuxi, China. His research interests include synthetic aperture radar (SAR) motion compensation, circular SAR, and high forward-squint SAR.



Shaohua Ye was born in Wuxi, China, in 1966. He received the bachelor's degree from Yangzhou Institute of Technology, Yangzhou, China, in 1987, the M.S. degree from Southeast University (SEU), Nanjing, China, in 1996, and the Ph.D. degree from Nanjing University of Aeronautics and Astronautics (NUAA), Nanjing, in 2002, all in electronic engineering.

From 1987 to 1993, he was a teacher at Wuxi Commercial School, Wuxi. In 1996, he joined the Radar and Avionics of AVIC I, Wuxi, where he is now a Professor. From 1998 to 1999, he was a Guest Scientist at the Institute of Radio Frequency Technology, German Aerospace Center (DLR), Oberphaffenhofen, Germany, working in the field of SAR raw data compression. His research interests include the research and development of advanced synthetic aperture radar (SAR) systems, radar imaging algorithms, and SAR/GMTI.



Daiyin Zhu (Member, IEEE) received the B.S. degree in electronic engineering from Southeast University, Nanjing, China, in 1996, and the M.S. and Ph.D. degrees in electronics from Nanjing University of Aeronautics and Astronautics (NUAA), Nanjing, in 1998 and 2002, respectively.

From 1998 to 1999, he was a Guest Scientist at the Institute of Radio Frequency Technology, German Aerospace Center (DLR), Oberpfaffenhofen, Germany, where he worked in the field of SAR interferometry. In 1998, he joined the Department of Electronic Engineering, NUAA, where he is currently a Professor. He has developed algorithms for several operational airborne synthetic aperture radar (SAR) systems. His research interests include radar imaging algorithms, SAR ground moving target indication (SAR/GMTI), SAR/ISAR autofocus techniques, SAR interferometry, and multiple-input multiple-output (MIMO) SAR signal processing.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 02:12:53 UTC from IEEE Xplore. Restrictions apply.